Interaction between fluids and Helfrich membranes

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I. INTRODUCTION

A. State of the art

A number of existing studies rely on the hypothesis that the bulk fluid surrounding the Helfrich membrane is inertialess. For instance, Arroyo et al. study the interaction between a Helfrich membrane and bulk fluid in the Stokes approximation, for simple membrane geometries that can be described by a single parameter [1]. Boedec and coauthors combined finite element (FE) and boundary element (BE) methods describe an Helfrich membrane interacting with an inertialess bulk fluid [2]. In there, the membrane surface rheology, such as viscous effects and flows induced by surface-tension gradients, is neglected. A Helfrich membrane coupled to a bulk fluid has been recently studied by Zhou and co-authors with BE and FE methods, under the assumption that the bulk fluid is inertialess and that Helfrich membrane is inextensible [3]. Barakat and Shaqfeh modelled vesicles in a circular tube by means of BE methods and lubrication theory, by neglecting Helfrich membrane rheology and describing bulk fluid as inertialess [4]. Finally, Rodrigues et al., studied the interaction between an inertialess, closed Helfrich membranes and bulk fluids with FE methods, by adopting a minimal description for the fluid, which is solely modelled as a volume constraint [5].

Other studies go beyond the inertialess approximation, and rely on simplifying assumptions for the Helfrich membrane. For instance, Bonito et al. propose a FE model for a Helfrich membrane interacting an bulk fluid, the latter being in the inertial regime [6, 7]. In this study Helfrich membrane is treated as a purely elastic body, characterized solely by bending rigidity, area and volume constraints—its fluid features, such as tangential and normal flows, are neglected. Laadhari extended the analyses above to the interaction between a Helfrich membrane and a non-Newtonian bulk fluid, in which membrane rheology is still neglected [8]. The interaction between a Helfrich membrane and a bulk fluid—both in the inertial regime—has been studied by Barrett and co-authors, [9], under the assumption that the M is perfectly inextensible. Such analysis is based on a fluidic approach, in which the Helfrich membrane does not obey its own, intrinsic rheology—see below—but it is advected by the surrounding bulk fluid. As a result, the membrane tangential and normal velocities are not determined as the result of the surfacial Navier-Stokes (NS) equations, but they match the respective components of the bulk fluid velocity.

In this study, we propose a novel FE study of the interaction between a bulk fluid (F) and a Helfrich membrane (M), based on the arbitrary Lagrangian-Eulerian (ALE) method [10, 11]. Our analysis includes a complete description of the physical features of both M and F, it may describe the inertial regime of both, and it accounts for a detailed description of M elasticity, rheology, and extensibility. Unlike fluidic approaches where the M dynamics is determined by the F, here both the flows and shape of M are determined by intrinsic physical evolution equations. In addition, our study can account for both closed and open surfaces, by leveraging the flexibility of its FE formulation [12].

We solve for the system rheology by implementing a splitting scheme [12, 13] for the NS equations, applied simultaneously to the M and F flow dynamics, and demonstrate it to be numerically stable. As for the membrane shape, we generalize the arc length (AL) gauge—previously proposed to describe the steady state of membrane tubes [14]—to the system dynamics, by means of a time-dependent gauge fixing. Overall, this implementation allows for handling strongly turbulent M and F flows, as well as large M deformations.

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The paper is structured as follows. In Section II we discuss the ALE method, and present two illustrative examples: The interaction between F and a rigid body (R) in Section II A, and that between F and E in Appendix B. In Section II C we present our approach for the interaction between F and M. In particular, in Sections II C 1 to II C 3, we discuss the splitting scheme and gauge-fixing procedure for the M, bulk fluid domain (D) and F, respectively. Finally, Section III is devoted to the discussion of the results, and future directions.

II. RESULTS

In all systems that we will consider, we will introduce the reference (r) and current (c) configuration [15–17], see Fig. 2. All quantities relative to the r and c configuration will be denoted by the superscript r and c, respectively. In the r configuration, the region occupied by F is described by Cartesian coordinates $\mathbf{x}_{\mathrm{r}}$. In the c configuration, the region occupied by F is described by Cartesian coordinates $\mathbf{x}_{\mathrm{c}}$.

The r configuration is related to the c one as follows [15], see Fig. 2. At time t, the point $\mathbf{x}_{\mathrm{c}}$ corresponds to a point $\mathbf{x}_{\mathrm{r}}$ in the r configuration through the deformation field [18]

$$\boldsymbol{u}_{\mathrm{D}}^{t}(\mathbf{x}_{\mathrm{r}}) \equiv \mathbf{x}_{\mathrm{c}} - \mathbf{x}_{\mathrm{r}}. \tag{1}$$

and the deformation-gradient tensor

$$F_{\alpha\beta}(\boldsymbol{u}) \equiv \delta_{\alpha\beta} + \frac{\partial u^{\alpha}}{\partial \mathrm{x}_{\mathrm{r}\,\beta}}. \tag{2}$$

In what follows, greek indices $\alpha, \beta, \cdots$ will be used to denote vectors and tensors in two-dimensional Euclidean space—their position as upper or lower indices is thus immaterial [16, 19].

We will characterize the mapping between $\mathbf{x}_{\mathrm{c}}$ and $\mathbf{x}_{\mathrm{r}}$ configurations in Eq. (1) with the fields

$$\mathbf{x}_{\mathrm{c}} = \boldsymbol{\varphi}^{t}(\mathbf{x}_{\mathrm{r}}), \tag{3}$$

$$\mathbf{x}_{\mathrm{r}} = \boldsymbol{\psi}^{t}(\mathbf{x}_{\mathrm{c}}). \tag{4}$$

We denote the F velocity and surface tension, the negative of the pressure, by $v_{\mathrm{F\,c}}(\mathbf{x}_{\mathrm{c}})$ and $\varsigma_{\mathrm{F}}^{\mathrm{c}}(\mathbf{x}_{\mathrm{c}})$, respectively, where the superscript c specifies that these fields depend on the coordinate $\mathbf{x}_{\mathrm{c}}$ in the c configuration.

For each field $\boldsymbol{v}_{\mathrm{F\,c}}(\mathbf{x}_{\mathrm{c}}), \varsigma_{\mathrm{F}}(\mathbf{x}_{\mathrm{c}})$ which depends on the coordinate $\mathbf{x}_{\mathrm{c}}$ in the c configuration, we introduce its counterpart in the r configuration:

$$\boldsymbol{v}_{\mathrm{F\,r}}(\mathbf{x}_{\mathrm{r}}, t) \equiv \boldsymbol{v}_{\mathrm{F\,c}}(\boldsymbol{\varphi}^{t}(\mathbf{x}_{\mathrm{r}}), t), \tag{5}$$

$$\varsigma_{\mathrm{F}}^{\mathrm{r}}(\mathbf{x}_{\mathrm{r}}, t) \equiv \varsigma_{\mathrm{F}}^{\mathrm{c}}(\boldsymbol{\varphi}^{t}(\mathbf{x}_{\mathrm{r}}), t). \tag{6}$$

The solution of the interaction dynamics between F and R, E or M is involved because of the presence of a moving boundary, cf. Figs. 2, 3 and 5, respectively. Following the ALE procedure, we will bridge between the Lagrangian description used to describe R, E or M, and the Eulerian description of F [15]. To achieve this, in what follows we will consider the equations of motion for R, E or M and F—which are naturally formulated the c configuration—and rewrite them in the r configuration.

A. Bulk fluid and rigid body

In this Section, we will discuss the interaction between a bulk fluid (F) and the simplest structure—a rigid body (R) which is only allowed to turn about one point. Only the main results will be presented here; details can be found in Appendices A 1 to A 3, respectively, and the temporal discretization is discussed in Appendix A 4.

As shown in Fig. 2, in the reference (r) configuration the axis of the ellipse, e.,g, R, lies parallel to the $\mathrm{x}_{\mathrm{c}\,1}$ axis, while in the current (c) configuration, such axis forms an angle θ (red arc) with respect to the $\mathrm{x}_{\mathrm{c}\,1}$ axis,

An example of the system dynamics is shown in Fig. 1, where we study the interaction between air (F) and an aerogel (R)—an ultra light solid material involving a large porous structure [20]. The channel dimension are $L = 2.2\,\mathrm{m}$, $h = 0.41\,\mathrm{m}$, which have been taken from the Finite Element Analysis Tools 2D (FEAT2D) DFG 2D-3 benchmark example for a fuid flow around a cylinder [21, 22], see Fig. 2. Given that the materials, such as F and R, are two dimensional, in what follows we will estimate their parameter values by considering their three-dimensional counterparts, and assume a unit thickness in the out-of-plane direction. The parameters for R have been estimated from the properties of air: $\rho_{\mathrm{F}} = 1\,\mathrm{Kg/m^2}$, $\eta_{\mathrm{F}} = 10^{-5}\,\mathrm{Pa\,m\,sec}$ [23]. The inflow velocity profile is chosen as a Poiseuille flow

$$\boldsymbol{v}_{\mathrm{F}\,\sqsubset}(\mathbf{x}_{\mathrm{c}}) = \frac{6 v_0}{h^2} \mathrm{x}_{\mathrm{c}\,2}(\mathrm{x}_{\mathrm{c}\,2} - h), \tag{7}$$

with $v_0 = 1\,\mathrm{m/sec}$. The mesh of the bulk fluid domain (D) deforms in time, and its rigidity is determined by the stiffness exponent ζ [15], which has been chosen to be $\zeta = 3$, see Section II C 2 for details. Finally, R is an ellipse with semi-axes $a = 0.1\,\mathrm{m}$, $b = 0.075\,\mathrm{m}$, center located at $\mathbf{x}_{\mathrm{r}} = (0.25\,\mathrm{m}, 0.2\,\mathrm{m})$, and moment of inertia $I = 10^{-3}\,\mathrm{Kg\,m}^2$, which corresponds to an aerogel density of $\sim 6\,\mathrm{Kg/m}^3$ [24]. As shown in Fig. A.1, R has an oscillatory behavior, whose period can be understood with scaling arguments—see Appendix A 5.

B. Bulk fluid and elastic body

In this Section, we will build on the analysis of Section II A and put F into interaction with a more complex physical object—an elastic body (E); only the main results will be presented here.

The r and c configurations are shown in Fig. 3. Here, the E boundary $\partial\Omega_{\bigcirc}$ may deform, but the inner E boundary $\partial\Omega_{\odot}$ is pinned to the $\mathrm{x}_{\mathrm{c}\,1}$, $\mathrm{x}_{\mathrm{c}\,2}$ plane. The equations of motion are reported in Appendix B 1, and the system dinamics is discussed in Appendix B 2.

An example of the coupled dynamics of F and E is shown in Fig. 4, where we show the interaction of a F with a hydrogel (E). Here, $\partial\Omega^{\mathrm{r}}_{\bigcirc}$ is an ellipse with semi-axes $a = 0.2\,\mathrm{m}$, $b = 0.1\,\mathrm{m}$ and center located at $\mathbf{x}_{\mathrm{r}} = (0.4\,\mathrm{m}, 0.2\,\mathrm{m})$, and $\partial\Omega^{\mathrm{r}}_{\odot}$ has radius $r = 0.0125\,\mathrm{m}$ and center located at $\mathbf{x}_{\mathrm{r}}$. The channel dimensions and F parameters have been taken from the FEAT2D DFG 2D-3 benchmark example for a fluid flow around a cylinder [21, 22]. The E density is $\rho_{\mathrm{E}} = 10^3\,\mathrm{Kg/m}^2$ [25]. The storage modulus, which approximates the shear modulus μ [26] is $\sim 10\,\mathrm{Kg/\,sec}^2$ [27]. Given a E Poisson ratio [16] of $\nu_{\mathrm{P}} \sim 0.49$ [28], we obtain $K \sim 5 \times 10^2\,\mathrm{Kg/\,sec}^2$ from the relation $K = 2\mu(1+\nu_{\mathrm{P}})/[3(2\nu_{\mathrm{P}}-1)]$ [16]. The F inflow velocity profile is a Poiseuille flow, Eq. (7), with $v_0 = 1\,\mathrm{m/\,sec}$. Finally, the D mesh stiffness exponent has been chosen to be $\zeta = 3$ [15].

C. Bulk fluid and Helfrich membrane

In this Section, we will build on the analysis of Appendix B, and complexify the structure of the elastic body with which F interacts by considering, instead of an elastic material, an M [14, 29].

For this problem, we will consider the geometry depicted in Fig. 5. A vertical channel with mirror symmetry about its mid axis is considered. The left boundary of the channel is $\partial\Omega_{|\square}$, and the right boundary $\partial\Omega_{\square|}$ lies on the symmetry axis. Fluid is injected from bottom to top on $\partial\Omega_{\underline{\square}}$. Here, M lies on the top boundary of F, $\partial\Omega_{\overline{\square}}$, which coincides with the M manifold $\partial\Omega_{\_\_}$. Finally, we denote by Ω the region enclosed by the boundaries, and all the quantities above will be considered in the r and c configurations.

The geometry of Fig. 5 may represent F being confined in a large reservoir, lying below the channel, where the channel is a vertical neck of the reservoir, covered on top by M.

1. *Helfrich membrane motion*

Both the physical nature and mathematical description of M is significantly more complex than that of E. First, unlike E, M has a fluid structure: its material elements can both flow tangentially and move normally to it. Such fluid behavior is suited to an Eulerian description, which then needs to be connected to the Lagrangian description used to describe D. Second, M is described by fourth-order partial differential equations (PDEs) [30]. From the mathematical, numerical and physical stand point, such PDEs are significantly more complex than the second-order PDEs which describe E [31]. Third, the geometrical description of M requires fixing an appropriate *gauge*. In fact, a general parametrization of M is given by two functions $\mathbf{x}_{\mathrm{c}}{}^{\mathrm{M}}(x^1)$. Both of these functions depend on the curvilinear coordinate x^1 which is defined in the r domain of M, $\partial\Omega^{\mathrm{r}}_{\_\_}$, see Fig. 5—note that here x^i differs from the Cartesian coordinate x^{α}. Despite this pair of functions, a single PDE determines the M shape [30]. To understand this redundancy, consider the stretching field $\nu > 0$, defined by

$$\nu^2 \equiv \boldsymbol{e}_1 \cdot \boldsymbol{e}_1, \tag{8}$$

$$\boldsymbol{e}_i \equiv \frac{\partial \mathbf{x}_{\mathrm{c}}{}^{\mathrm{M}}}{\partial x^i}. \tag{9}$$

It is then easy to realize that two parametrizations with different νs may yield the same physical shape. Such redundancy may be removed by a gauge fixing, i.e., by adding a constraint on $\mathbf{x}_{\mathrm{c}}{}^{\mathrm{M}}$, or its derivatives. Two common gauge choices are the Monge [32, 33], or the arc-length parametrization [14], in which one sets

$$\nu = 1. \tag{10}$$

The arc-length gauge has been used to describe, for instance, the steady state for membrane tubules [14]. However, while a static constraint such as (10) is suited to static shapes, it cannot describe the M dynamics, such as the one of M interacting with F. In fact, as M is deformed, it will also

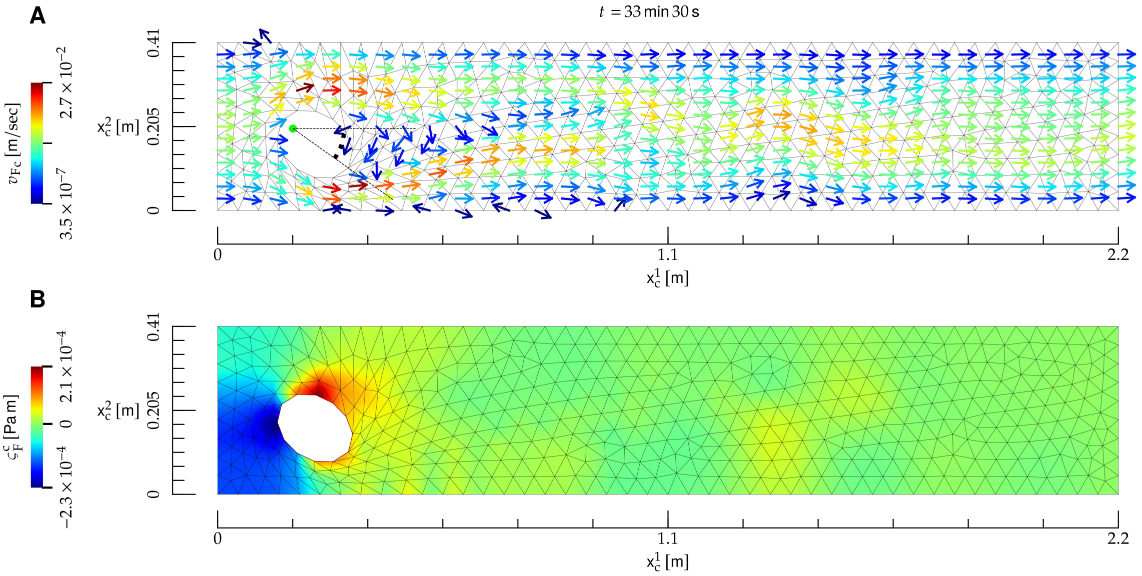

FIG. 1: Interaction between a bulk fluid (F) and a rigid rigid body (R), whose material properties roughly correspond to air (F) and an aerogel (R). **A**) Temporal snapshot of the mesh (black lines) and velocity field (arrows) at the current time t, shown on top. The direction of the velocity field is indicated by the arrows, and its norm by their color; the direction of arrows with velocity close to zero is not defined. Here, R (ellipse) is allowed to pivot about its left focal point (red dot), and the related angle with respect to the $\mathsf{x}_{\mathrm{c}\,1}$ direction is denoted by a dashed arc. Here, the Reynolds number is $R \sim 2 \times 10^2$. **B**) Color map of the surface tension at the current instant of time t.

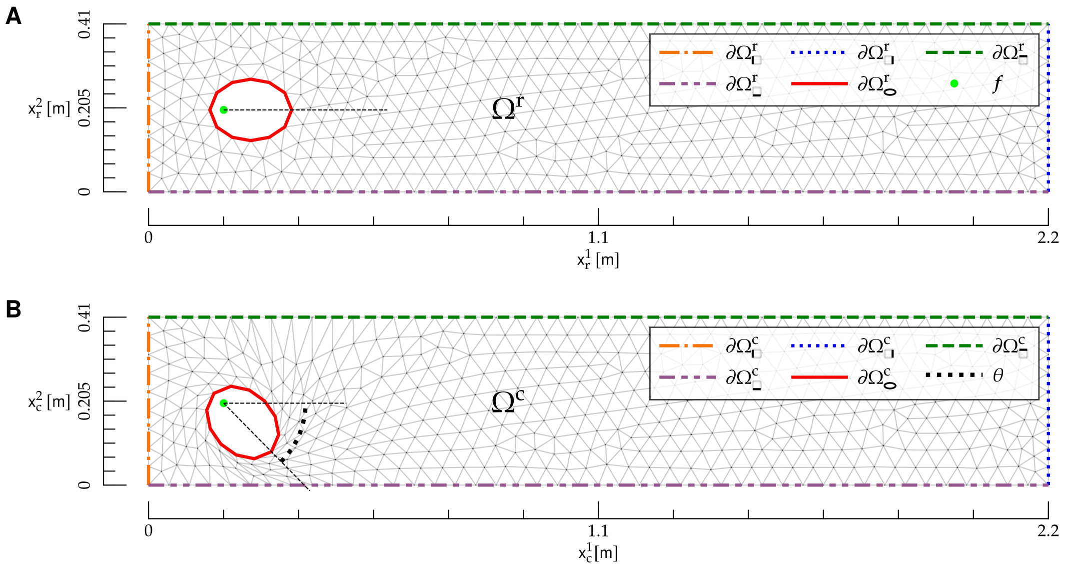

FIG. 2: Reference (r) and current (c) configuration for the interaction between a bulk fluid and a rigid body (R). **A**) r configuration. The region Ω^{r} is given by the rectangle $0 \leq \mathsf{x}_{\mathrm{r}\,1} \leq L$, $0 \leq \mathsf{x}_{\mathrm{r}\,2} \leq h$ with the elliptical hole (R), and is delimited by the mesh (gray lines). Boundaries in the r configuration are denoted by $\partial\Omega^{\mathrm{r}}_{\sqsubset}$, $\partial\Omega^{\mathrm{r}}_{\sqsupset}$, $\partial\Omega^{\mathrm{r}}_{\sqcap}$, $\partial\Omega^{\mathrm{r}}_{\sqcup}$ and $\partial\Omega^{\mathrm{r}}_{\circ}$ (colored dashed lines). The ellipse focal point $\boldsymbol{f}$ (light green dot) is also shown. **B**) c configuration. The region Ω^{c} is delimited by the mesh (gray lines). Boundaries in the c configuration are denoted by $\partial\Omega^{\mathrm{c}}_{\sqsubset}$, $\partial\Omega^{\mathrm{c}}_{\sqsupset}$, $\partial\Omega^{\mathrm{c}}_{\sqcap}$, $\partial\Omega^{\mathrm{c}}_{\sqcup}$ and $\partial\Omega^{\mathrm{c}}_{\circ}$ (colored dashed lines), and the rotational angle θ of R is shown as an arc (black dotted line). The Cartesian coordinates $\mathbf{x}_{\mathrm{c}}$ are also shown.

stretch: as a result, Eq. (10) must be replaced by

a time-dependent gauge—see below.

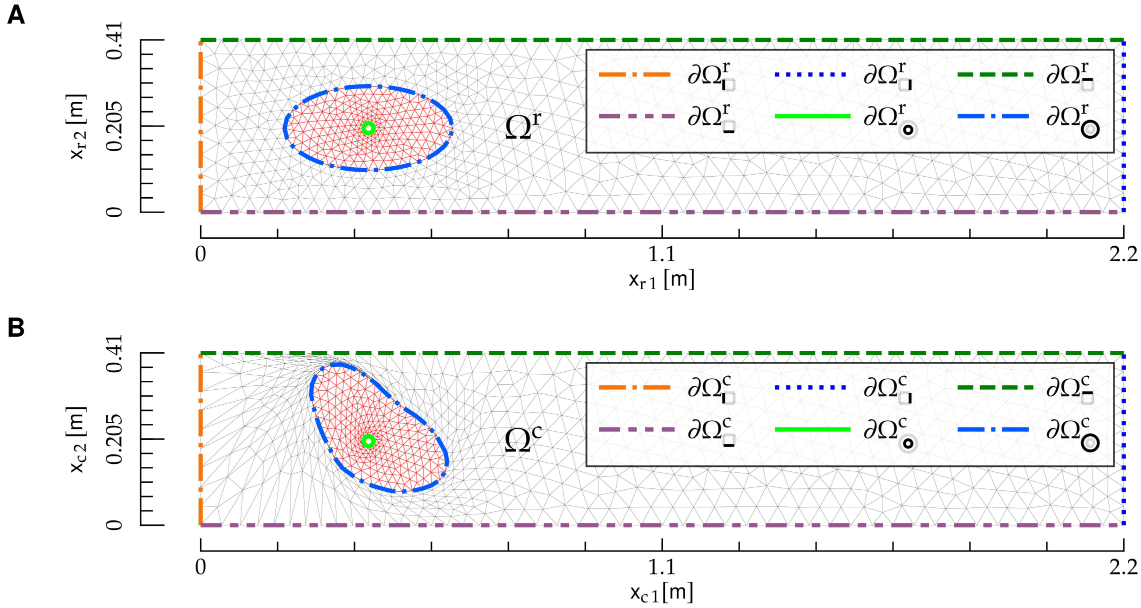

FIG. 3: Reference and current configuration for the interaction between a bulk fluid and an elastic body. The figures follows the same notation as Fig. 2.

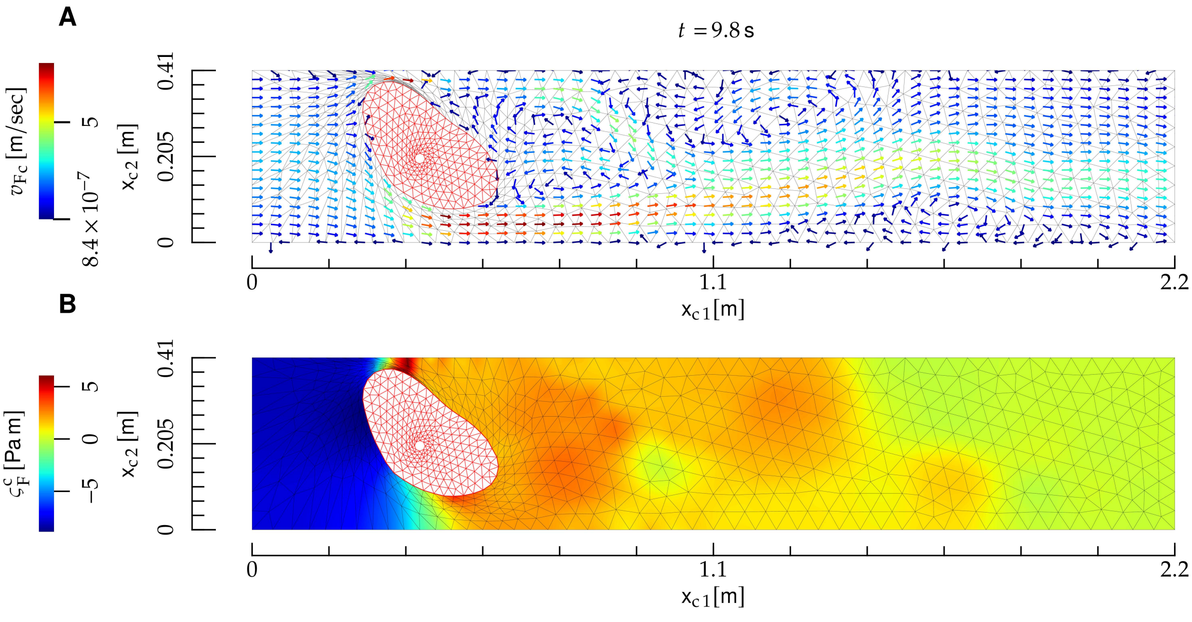

FIG. 4: Interaction between a bulk fluid and an elastic body (E) (a hydrogel). The notation is the same as in Fig. 1, and the elastic body is depicted as a red mesh. Here, the Reynolds number is $R \sim 2 \times 10^2$. The E deformation is not symmetric under the mirror symmetry $\mathrm{x}_{\mathrm{c}\,2} \to h - \mathrm{x}_{\mathrm{c}\,2}$, where h is the height of the F domain, because the center of E is not located at $\mathrm{x}_{\mathrm{c}\,2} = h/2$.

We describe M with the Lagrangian approach, along the lines of the description of D in Section II A and E in Appendix B. The M displacement field is

$$\boldsymbol{u}_\mathrm{M}(x^1) \equiv \mathbf{x}_\mathrm{c}{}^\mathrm{M}(x^1) - \mathbf{x}_\mathrm{r}{}^\mathrm{M}(x^1), \tag{11}$$

where we choose the r configuration as a straight line

$$\mathbf{x}_\mathrm{r}{}^\mathrm{M}(x^1) \equiv (x^1, h),\ 0 \leq x^1 \leq L, \tag{12}$$

which is shown in Fig. 5A. In addition, we introduce the angle ψ formed by the tangent to M with the

x^1 axis [14], defined by

$$e_1^1 = \nu\cos\psi, \tag{13}$$
$$e_1^2 = -\nu\sin\psi. \tag{14}$$

Finally, the M metric tensor is defined by [19]

$$g_{ij} \equiv \boldsymbol{e}_i\cdot\boldsymbol{e}_j. \tag{15}$$

In what follows, latin indices $i, j, k, \cdots$ will be used to denote vectors and tensors related to the M manifold; their position as upper and lower indexes is thus important [19]. Because the differential manifold of M is one dimensional, indexes $i, j, k, \cdots$ may take only one value, i.e., $i = j = k = \cdots = 1$. However, in formal relations such as Eq. (15), we will use the general notation $i, j, k, \cdots$ for the sake of generality, and for potential extensions of such relations to two-dimensional manifolds [33].

In order to work out the dynamical equations for M, let us derive the force exerted by F on M. The force exerted on a line element dl_{c} of M is [18]

$$f_{\mathrm{F}\,\alpha}(\sigma^{\mathrm{c}}_{\mathrm{F}}, \nu, \psi) = -\sigma^{\mathrm{c}}_{\mathrm{F}\,\alpha\beta}\hat{n}^{\mathrm{c}\,\beta}(\nu,\psi), \tag{16}$$

where

$$\sigma^{\mathrm{c}}_{\mathrm{F}\,\alpha\beta} \equiv \varsigma^{\mathrm{c}}_{\mathrm{F}}\delta_{\alpha\beta} + \eta_{\mathrm{F}}\left(\frac{\partial v_{\mathrm{F}}{}^{\alpha}_{\mathrm{c}}}{\partial \mathsf{x}_{\mathrm{c}\,\beta}} + \frac{\partial v_{\mathrm{F}}{}^{\beta}_{\mathrm{c}}}{\partial \mathsf{x}_{\mathrm{c}\,\alpha}}\right) \tag{17}$$

is the F stress tensor [18] and $\hat{n}^{\mathrm{c}}$ the unit normal to $\partial\Omega^{\mathrm{c}}_{\square}$ pointing outside Ω^{c}, see Fig. 5. In Eq. (16), $\sigma^{\mathrm{c}}_{\mathrm{F}}$ is evaluated at the c coordinate $\mathbf{x}_{\mathrm{c}}{}^{\mathrm{M}} \in \partial\Omega^{\mathrm{c}}_{\square}$ corresponding to x^1. The components of the force (16) tangential and normal to M are [33]

$$f^i_{\mathrm{F\,t}}(\sigma^{\mathrm{c}}_{\mathrm{F}}, \nu, \psi) = g^{ij}(\nu)e^{\alpha}_j(\nu,\psi) f_\alpha(\sigma^{\mathrm{c}}_{\mathrm{F}}, \nu, \psi) \tag{18}$$
$$f_{\mathrm{F\,n}}(\sigma^{\mathrm{c}}_{\mathrm{F}}, \nu, \psi) = \hat{n}^{\mathrm{c}\,\alpha}(\nu,\psi) f_\alpha(\sigma^{\mathrm{c}}_{\mathrm{F}}, \nu, \psi), \tag{19}$$

where in Eq. (18) we used the feature that the metric tensor depends on ν only, see Eqs. (8) and (15).

The equations of motion for M are [1, 31, 34]

$$\nabla_i v^i_{\mathrm{M}} - 2Hw_{\mathrm{M}} = 0, \tag{20}$$
$$\rho_{\mathrm{M}}\left(\frac{\partial v^i_{\mathrm{M}}}{\partial t} + v^j_{\mathrm{M}}\nabla_j v^i_{\mathrm{M}} - 2v^j_{\mathrm{M}}w_{\mathrm{M}}\, b^i_j - w_{\mathrm{M}}\nabla^i w_{\mathrm{M}}\right) = \nabla^i\varsigma_{\mathrm{M}} + f^i_{\eta_{\mathrm{M}}\,\mathrm{t}} + f^i_{\mathrm{F\,t}}, \tag{21}$$
$$\rho_{\mathrm{M}}\left[\frac{\partial w_{\mathrm{M}}}{\partial t} + v^i_{\mathrm{M}}(v^j_{\mathrm{M}}b_{ji} + \nabla_i w_{\mathrm{M}})\right] = 2\varsigma_{\mathrm{M}}H + f_\kappa + f_{\eta_{\mathrm{M}}\,\mathrm{n}} + f_{\mathrm{F\,n}}, \tag{22}$$
$$\frac{\partial \boldsymbol{u}_{\mathrm{M}}}{\partial t} = w_{\mathrm{M}}\hat{n}^{\mathrm{c}}, \tag{23}$$
$$\frac{\partial u^1_{\mathrm{M}}}{\partial x^1} = -1 + \nu\cos\psi, \tag{24}$$
$$\frac{\partial u^2_{\mathrm{M}}}{\partial x^1} = -\nu\sin\psi, \tag{25}$$

and they hold for $x^1 \in \Omega^{\mathrm{r}}_{\_}$. Here, v^i_{M} and w are the M tangential and normal velocity, respectively, ∇ is the covariant derivative related to the Levi-Civita connection associated to g, Δ_{LB} the Laplace-Beltrami operator, b the second fundamental form, and H and K the mean and Gaussian curvature, respectively. We refer to [1, 19, 31, 33] for details on the differential-geometry formulation, and we will now discuss each line in Eqs. (20) to (25). Equation (20) enforces mass conservation [18]. Equations (21) and (22) are the tangential and normal components of the Navier-Stokes (NS) equations, respectively. The first term in the right-hand side (RHS) of Eq. (21) represents the force on membrane elements induced by gradients of line tension, while the first term in the RHS of Eq. (22) constitutes Laplace pressure [18, 35]. The other terms read

$$f_\kappa(\nu,\psi) \equiv -2\kappa[\nabla_i\nabla^i H + 2H(H^2 - K)], \tag{26}$$
$$f^i_{\eta_{\mathrm{M}}\,\mathrm{t}}(v_{\mathrm{M}}, w_{\mathrm{M}}, \nu, \psi) \equiv \eta_{\mathrm{M}}[-\Delta_{\mathrm{LB}}v^i_{\mathrm{M}} - 2\left(b^{ij} - 2\,H\,g^{ij}\right)\nabla_j w_{\mathrm{M}} + 2Kv^i_{\mathrm{M}}], \tag{27}$$
$$f_{\eta_{\mathrm{M}}\,\mathrm{n}}(v_{\mathrm{M}}, w_{\mathrm{M}}, \nu, \psi) \equiv 2\eta_{\mathrm{M}}[(\nabla^i v^j_{\mathrm{M}})b_{ij} - 2w_{\mathrm{M}}(2H^2 - K)]. \tag{28}$$

In Eq. (26), f_κ is represents the resistance of M to bending, and it is derived from Helfrich model [29], while $f_{\eta_{\mathrm{M}}\, t}$ and $f_{\eta_{\mathrm{M}}\, \mathrm{n}}$ in Eqs. (27) and (28) are, respectively, the tangential and normal component of the viscous force resulting from in-membrane flows [1]. Equation (23) is the time-evolution equation for the M shape, which evolves according to the normal $\hat{n}^{\mathrm{c}}$, and it has been obtained from Eqs. (11) and (12). We observe that the M shape velocity, $\frac{\partial \boldsymbol{u}_{\mathrm{M}}}{\partial t}$, coincides with the normal displacement velocity of M, but is it independent of the tangential flow velocity $\boldsymbol{v}_{\mathrm{M}}$ of M elements along the shape. This implies that the M shape and F motions are conforming [15], but that the tangential part of the F velocity at M need not coincide with $\boldsymbol{v}_{\mathrm{M}}$, thus allowing for a slip [18]. Finally, Eqs. (24) and (25) re-express the definitions of ν and ψ, and result from Eqs. (9) and (11) to (14), and ρ_{M}, ς_{M}, η_{M} and κ are, respectively, the density, line tension, viscosity, and bending rigidity of M.

We choose the following boundary conditions (BCs) for Eqs. (20) to (25):

$$v^i_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{29}$$

$$v^i_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{30}$$

$$w_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{31}$$

$$\nabla_i w_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{32}$$

$$\varsigma_{\mathrm{M}} = \varsigma_{\mathrm{M}\,\bullet\!-} \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{33}$$

$$\boldsymbol{u}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{34}$$

$$u^1_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{35}$$

$$\frac{\partial u^2_{\mathrm{M}}}{\partial x^1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}. \tag{36}$$

Equations (29), (31) and (34) enforce the condition that the left extremity of M, $\partial\Omega_{\bullet\!-}$, is clamped on the channel wall $\partial\Omega_{|\square}$. Equations (30), (32), (35) and (36) enforce the mirror symmetry about $\partial\Omega_{\square|}$. Namely, Eq. (30) ensures that the v_{M} vector field vanishes on $\partial\Omega_{-\!\bullet}$, because a nonzero value of this field on $\partial\Omega_{\square|}$ would break the axial symmetry. The condition (35) follows the same lines. In addition, the symmetry requires that scalar fields and vertical components of vector fields in Ω have zero slope with respect to x^1 at $\partial\Omega_{-\!\bullet}$, see Eqs. (32) and (36). Finally, Eq. (33) fixes the M tension on $\partial\Omega_{\bullet\!-}$.

2. *Bulk fluid domain motion*

We will determine $\boldsymbol{u}_{\mathrm{D}}$ as the solution of a boundary-value problem (BVP) given by the following elastic model [15, 36]

$$\frac{\partial S_{\mathrm{D}\,\alpha\beta}(\boldsymbol{u}_{\mathrm{D}})}{\partial \mathsf{x}_{\mathrm{r}\,\beta}} = 0 \text{ in } \Omega^{\mathrm{r}}, \tag{37}$$

where the elastic stress tensor S_{D} is given by

$$S_{\mathrm{D}\,\alpha\beta} \equiv F_{\alpha\gamma} T_{\gamma\beta}, \tag{38}$$

where F is defined in Eq. (2), and

$$\begin{aligned} T_{\alpha\beta} \equiv & K(\boldsymbol{u}_{\mathrm{D}}) E_{\gamma\gamma}\delta_{\alpha\beta} + \\ & 2\mu(\boldsymbol{u}_{\mathrm{D}}) \left(E_{\alpha\beta} - \frac{1}{2}\delta_{\alpha\beta}E_{\gamma\gamma} \right), \end{aligned} \tag{39}$$

$$E_{\alpha\beta} \equiv \frac{1}{2}\left(F_{\gamma\alpha}F_{\gamma\beta} - \delta_{\alpha\beta}\right), \tag{40}$$

$$K(\boldsymbol{u}) \equiv \frac{1}{[\det(F(\boldsymbol{u}))]^{\zeta}}, \tag{41}$$

$$\mu(\boldsymbol{u}) \equiv \frac{1}{[\det(F(\boldsymbol{u}))]^{\zeta}}. \tag{42}$$

In Eqs. (41) and (42), we included an additional dependence of bulk modulus K and the shear modulus μ on the deformation-gradient tensor (2). In the absence of this dependence, the elastic model above would be unstable under local compression [15]. In fact, the model's Lagrangian would not penalize configurations with small $\det F$. The inclusion of the dependence (41) and (42) on $\det F$ allows for penalizing these configurations—the larger the exponent $\zeta > 0$, the stronger the penalty. As a result, this dependence makes the model stable under mesh compression, such as the one shown below R in Fig. 2B.

The equation of motion for $\dot{\boldsymbol{u}}_{\mathrm{D}}$ is obtained by deriving Eq. (37) with respect to time, and using the fact that r coordinates are time independent:

$$\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}\,\beta}} \frac{\mathrm{d}S_{\mathrm{D}\,\alpha\beta}(\boldsymbol{u}_{\mathrm{D}})}{\mathrm{d}t} = 0 \text{ in } \Omega^{\mathrm{r}}. \tag{43}$$

We impose the following BCs for Eqs. (37) and (43):

$$\boldsymbol{u}_{\mathrm{D}} = \boldsymbol{0} \text{ on } \partial\Omega^{\mathrm{r}}_{|\square} \cup \partial\Omega^{\mathrm{r}}_{\underline{\square}}, \tag{44}$$

$$u^1_{\mathrm{D}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{45}$$

$$\frac{\partial u^2_{\mathrm{D}}}{\partial \mathsf{x}_{\mathrm{r}\,1}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{46}$$

$$\boldsymbol{u}_{\mathrm{D}} = \boldsymbol{u}_{\mathrm{M}} \text{ on } \partial\Omega^{\mathrm{r}}_{\overline{\square}}, \tag{47}$$

$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \boldsymbol{0} \text{ on } \partial\Omega^{\mathrm{r}}_{|\square} \cup \partial\Omega^{\mathrm{r}}_{\underline{\square}}, \tag{48}$$

$$\dot{u}^1_{\mathrm{D}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{49}$$

$$\frac{\partial \dot{u}^2_{\mathrm{D}}}{\partial \mathsf{x}_{\mathrm{r}\,1}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{50}$$

$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \dot{\boldsymbol{u}}_{\mathrm{M}} \text{ on } \partial\Omega^{\mathrm{r}}_{\overline{\square}}. \tag{51}$$

Equation (44) ensures that D is not deformed on $\partial\Omega^{\mathrm{r}}_{|\square} \cup \partial\Omega^{\mathrm{r}}_{\underline{\square}}$, cf. Eq. (A.18). Equations (45) and (46)

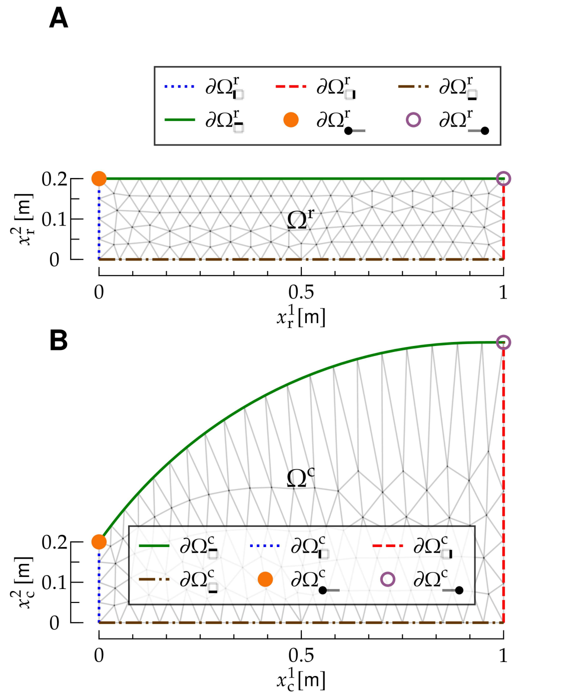

FIG. 5: Reference and current configuration for the interaction between a bulk fluid (F) and a Helfrich membrane (M). The Figure follows the same notation as Fig. 2, and it represents a vertical channel with mirror symmetry about its mid axis, which coincides with $\partial\Omega_{\square|}$; the right, mirrored part of the channel is not shown. The top edge of the F mesh, $\partial\Omega^{\mathrm{r}}_{\overline{\square}}$, coincides with the one-dimensional manifold $\Omega^{\mathrm{r}}_{\overline{}}$ of M.

follow from axial symmetry, and Eq. (47) from the requirement that D and M deformations stay conforming [15]. Finally, the BCs for $\dot{\boldsymbol{u}}_{\mathrm{D}}$, Eqs. (48) to (51), are derived along the same lines as Eqs. (44) to (47).

We observe that Eqs. (38) to (42) imply that Eqs. (37) and (43) contains up to second derivatives of $\boldsymbol{u}_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}$. As a result, the BCs (44) to (51) uniquely determine the solution of the BVP (37) and (43) [37].

3. *Bulk fluid motion*

We will now discuss the F equations of motion, and present them first in c configuration coordinates, and then transform them in r coordinates.

a. Equations of motion in current configuration coordinates In c coordinates, the F equations of motion are the NS and continuity equations [18]:

$$\rho_{\mathrm{F}}\left(\partial_t {v_{\mathrm{F}}}_{\mathrm{c}}^{\alpha} + {v_{\mathrm{F}}}_{\mathrm{c}}^{\beta}\frac{\partial {v_{\mathrm{F}}}_{\mathrm{c}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{c}\,\beta}}\right) = \frac{\partial {\sigma^{\mathrm{c}}_{\mathrm{F}}}_{\alpha\beta}}{\partial \mathsf{x}_{\mathrm{c}\,\beta}}, \tag{52}$$

$$\frac{\partial {v_{\mathrm{F}}}_{\mathrm{c}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{c}\,\alpha}} = 0, \tag{53}$$

and they hold for $\mathbf{x}_{\mathrm{c}} \in \Omega^{\mathrm{c}}$. Equations (52) and (53) describe the F motion in a flat, two-dimensional geometry [18]—an example of the corresponding NS equations on a curved geometry is given by Eq. (21).

We enforce the following BCs for Eqs. (52) and (53):

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}} = \boldsymbol{v}_{\mathrm{F}\,\underline{\square}} \text{ on } \partial\Omega^{\mathrm{c}}_{\underline{\square}}, \tag{54}$$

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}} = \boldsymbol{0} \text{ on } \partial\Omega^{\mathrm{c}}_{|\square}, \tag{55}$$

$${v_{\mathrm{F}}}_{\mathrm{c}}^{1} = 0 \quad \text{on } \partial\Omega^{\mathrm{c}}_{\square|}, \tag{56}$$

$$\frac{\partial {v_{\mathrm{F}}}_{\mathrm{c}}^{2}}{\partial \mathsf{x}_{\mathrm{c}\,2}} = 0 \quad \text{on } \partial\Omega^{\mathrm{c}}_{\square|}, \tag{57}$$

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}(\mathbf{x}_{\mathrm{c}}) = \left.\frac{\mathrm{d}\boldsymbol{u}_{\mathrm{M}}(x^1)}{\mathrm{d}t}\right|_{x^1 = [\boldsymbol{\psi}^t(\mathbf{x}_{\mathrm{c}})]^1} \quad \text{on } \partial\Omega^{\mathrm{c}}_{\overline{\square}}, \tag{58}$$

$$\varsigma^{\mathrm{c}}_{\mathrm{F}} = \varsigma_{\mathrm{F}\,\underline{\square}} \text{ on } \partial\Omega^{\mathrm{c}}_{\underline{\square}}. \tag{59}$$

According to Eq. (54), F is injected from the bottom edge of Ω^{c}, $\partial\Omega^{\mathrm{c}}_{\underline{\square}}$, while Eq. (55) is a no-slip BC on the left wall $\partial\Omega^{\mathrm{c}}_{|\square}$, and Eqs. (56) and (57) result from symmetry. Equation (58) ensures that the motion of F is conforming to that of M [15]: In fact, Eq. (23) ensures that M only moves normally to its profile, and Eq. (58) thus ensures that F stays in contact with M throughout its motion. Equation (58) ensures that the velocity of F and of the M shape coincide, allowing for a slip between the tangential flow of M elements and F material elements, see Section II C 1 and Eq. (23) in there. Finally, Eq. (59) stems from the presence of the reservoir, in which the F tension is equal to $\varsigma_{\mathrm{F}\,\underline{\square}}$: At the boundary $\partial\Omega^{\mathrm{c}}_{\underline{\square}}$ between the channel and the reservoir, the F tension matches the reservoir's.

b. Equations of motion in reference configuration coordinates We will now rewrite the equations of motion and the BCs of Section II C 3 a in terms of $\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}$, $\varsigma^{\mathrm{r}}_{\mathrm{F}}$, and the r coordinate $\mathbf{x}_{\mathrm{r}}$. The equations of motion (52) and (53) yield, respectively,

FIG. 6: Interaction between a bulk fluid (F) and a Helfrich membrane (M). Here, F is air, and M an elastomer, and the F Reynolds number is $R \sim 200$. **A**) Reference (r) and current (c) configuration of M (red and green curve, respectively), and M displacement field $\boldsymbol{u}_{\rm M}$ (black arrows). **B**) Stretch field of M. **C**) Tangent angle of M. **D**) and **E**): F velocity and tension, respectively; the notation is the same as in Fig. 1. **F**), **G**) and **H**): Tangential velocity, normal velocity and tension of M, respectively. All panels refer to the same instant of time, shown on top, and they display also the deformed mesh (gray lines).

$$\rho_{\rm F} \left\{ \frac{\partial {v_{\rm F}}_{\rm r}^{\alpha}(\mathsf{x}_{\rm r}, t)}{\partial t} + \left[{v_{\rm F}}_{\rm r}^{\gamma}(\mathsf{x}_{\rm r}, t) - \frac{{\rm d}\varphi^{\gamma\, t}(\mathsf{x}_{\rm r})}{{\rm d}t} \right] G_{\beta\gamma}(\boldsymbol{u}_{\rm D}^{t}(\mathsf{x}_{\rm r})) \frac{\partial {v_{\rm F}}_{\rm r}^{\alpha}(\mathsf{x}_{\rm r}, t)}{\partial {\mathsf{x}_{\rm r}}_{\beta}} \right\} = \tag{60}$$

$$G_{\beta\alpha}(\boldsymbol{u}_{\rm D}^{t}(\mathsf{x}_{\rm r})) \frac{\partial \varsigma_{\rm F}^{\rm r}(\mathsf{x}_{\rm r}, t)}{\partial {\mathsf{x}_{\rm r}}_{\beta}} + \eta_{\rm F}\, G_{\delta\beta}(\boldsymbol{u}_{\rm D}^{t}(\mathsf{x}_{\rm r})) \frac{\partial}{\partial {\mathsf{x}_{\rm r}}_{\delta}} \left[G_{\gamma\beta}(\boldsymbol{u}_{\rm D}^{t}(\mathsf{x}_{\rm r})) \frac{\partial {v_{\rm F}}_{\rm r}(\mathsf{x}_{\rm r}, t)}{\partial {\mathsf{x}_{\rm r}}_{\gamma}} \right],$$

$$G_{\beta\alpha}(\boldsymbol{u}_{\rm D}^{t}(\mathsf{x}_{\rm r})) \frac{\partial {v_{\rm F}}_{\rm r}^{\alpha}(\mathsf{x}_{\rm r}, t)}{\partial {\mathsf{x}_{\rm r}}_{\beta}} = 0, \tag{61}$$

which hold in $\Omega^{\rm r}$. In Eqs. (60) and (61) the tensor G is given by the inverse of F [15]:

$$G_{\alpha\beta}(\boldsymbol{u}) \equiv [F(\boldsymbol{u})]^{-1}_{\alpha\beta}. \tag{62}$$

4. Temporal discretization

To numerically solve for the dynamics in a time span $0 \leq t \leq T$, we discretize time by setting $\Delta t \equiv T/N$, $t_n \equiv n\, \Delta t$, with $n = 0, 1, \cdots, N$.

We combine the Crank Nicolson (CN) discretization method for NS equations [12, 38] with the arbitrary Lagrangian-Eulerian (ALE) formulation, by discretizing the dynamical equations for M, D and F. The dynamics of the system is obtained as follows.

Following the CN temporal-discretization scheme [38], we define fields at staggered, integer and semi-integer time steps [12]: For example, we set

$$\boldsymbol{v}_{{\rm F}\,{\rm r}}{}^{n}(\mathsf{x}_{\rm r}) \equiv \boldsymbol{v}_{{\rm F}\,{\rm r}}(\mathsf{x}_{\rm r}, t_n), \tag{63}$$

$$\varsigma_{\rm F}^{{\rm r}\; n-1/2}(\mathsf{x}_{\rm r}) \equiv \varsigma_{\rm F}^{\rm r} \left(\mathsf{x}_{\rm r}, \frac{t_n + t_{n-1}}{2} \right), \tag{64}$$

and similarly for other F and M fields.

a. Helfrich membrane We discretize Eqs. (20) to (25) as follows

$$\nabla_i^{n-1/2} v_{\rm M}^{i} - 2 H^{n-1/2} w_{\rm M}^{n} = 0, \tag{65}$$

$$\rho_{\rm M} \Big(\frac{v_{\rm M}^{n,\, i} - v_{\rm M}^{n-1,\, i}}{\Delta t} + \frac{3}{2} v_{\rm M}^{n-1,\, j} \nabla_j^{n-1/2} v_{\rm M}^{n-1,\, i} - \frac{1}{2} v_{\rm M}^{n-2,\, j} \nabla_j^{n-1/2} v_{\rm M}^{n-2,\, i} - 2 v_{\rm M}^{n-1,\, j} w_{\rm M}^{n-1} b^{n-1/2,\, i}{}_{j} - \tag{66}$$

$$w_{\rm M}^{n-1} \nabla^{n-1/2,\, i} w_{\rm M}^{n-1} \Big) =$$

$$\nabla^{n-1/2,\, i} {\varsigma_{\rm M}}^{n-1/2} + f^{i}_{\eta_{\rm M}\, {\rm t}} \left(\frac{{\boldsymbol{v}_{\rm M}}^{n} + {\boldsymbol{v}_{\rm M}}^{n-1}}{2}, \frac{w_{\rm M}^{n} + w_{\rm M}^{n-1}}{2}, \nu^{n-1/2}, \psi^{n-1/2} \right) +$$

$$f^{i}_{\rm F\, t} (\boldsymbol{\sigma}_{\rm F}^{\rm r} ({\boldsymbol{v}_{\rm F}}_{\rm r}{}^{n-1}, \varsigma_{\rm F}^{{\rm r}\; n-3/2}, \boldsymbol{u}_{\rm D}^{n-1}), \nu^{n-1/2}, \psi^{n-1/2}),$$

$$\rho_{\rm M} \left(\frac{w_{\rm M}^{n} - w_{\rm M}^{n-1}}{\Delta t} + v_{\rm M}^{n-1,\, i} v_{\rm M}^{n-1,\, j} b_{ji}^{n-1/2} + \frac{3}{2} v_{\rm M}^{n-1,\, i} \nabla_i^{n-1/2} w_{\rm M}^{n-1} - \frac{1}{2} v_{\rm M}^{n-2,\, i} \nabla_i^{n-1/2} w_{\rm M}^{n-2} \right) = \tag{67}$$

$$f_{\kappa}(\nu^{n-1/2}, \psi^{n-1/2}) + 2 {\varsigma_{\rm M}}^{n-1/2} H^{n-1/2} + f_{\eta_{\rm M}\, {\rm n}} \left(\frac{{\boldsymbol{v}_{\rm M}}^{n} + {\boldsymbol{v}_{\rm M}}^{n-1}}{2}, \frac{w_{\rm M}^{n} + w_{\rm M}^{n-1}}{2}, \nu^{n-1/2}, \psi^{n-1/2} \right) +$$

$$f_{\mathrm{F\,n}}(\boldsymbol{\sigma}^{\mathrm{r}}_{\mathrm{F}}(\boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1}, \varsigma^{\mathrm{r}}_{\mathrm{F}}{}^{n-3/2}, \boldsymbol{u}^{n-1}_{\mathrm{D}}), \nu^{n-1/2}, \psi^{n-1/2}),$$

$$\frac{\boldsymbol{u}^{n-1/2}_{\mathrm{M}} - \boldsymbol{u}^{n-3/2}_{\mathrm{M}}}{\Delta t} = w^{n-1}_{\mathrm{M}} \hat{n}^{\mathrm{c}\, n-1/2}, \tag{68}$$

$$\frac{\partial u^{n-1/2,\,1}_{\mathrm{M}}}{\partial x^1} = -\,1 + \nu^{n-1/2} \cos \psi^{n-1/2}, \tag{69}$$

$$\frac{\partial u^{n-1/2,\,2}_{\mathrm{M}}}{\partial x^1} = -\,\nu^{n-1/2} \sin \psi^{n-1/2}. \tag{70}$$

In Eqs. (66) and (67), $\sigma^{\mathrm{r}}_{\mathrm{F}}$ denotes the F stress tensor in the r configuration:

$$\begin{aligned}\sigma^{\mathrm{r}\,t}_{\mathrm{F}\,\alpha\beta}(\mathbf{x}_{\mathrm{r}}) \equiv& \sigma^{\mathrm{c}}_{\mathrm{F}\,\alpha\beta}(\boldsymbol{\varphi}^t(\mathbf{x}_{\mathrm{r}})) \\ =& \varsigma^{\mathrm{r}}_{\mathrm{F}}(\mathbf{x}_{\mathrm{r}}, t)\delta_{\alpha\beta} + \eta_{\mathrm{F}} \Big[G_{\gamma\beta}(\boldsymbol{u}^t_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}})) \frac{\partial {v_{\mathrm{F}}}^{\alpha}_{\mathrm{c}}}{\partial \mathrm{x}_{\mathrm{r}\,\gamma}} + \\ & G_{\gamma\alpha}(\boldsymbol{u}^t_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}})) \frac{\partial {v_{\mathrm{F}}}^{\beta}_{\mathrm{c}}}{\partial \mathrm{x}_{\mathrm{r}\,\gamma}} \Big].\end{aligned} \tag{71}$$

Proceeding along the same lines, the discrete form of the BCs (29) to (36) reads, respectively,

$$v^{n,\,i}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet-}, \tag{72}$$

$$v^{n,\,i}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\bullet}, \tag{73}$$

$$w^{n}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet-}, \tag{74}$$

$$\nabla^{n-1/2}_i w^{n}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\bullet}, \tag{75}$$

$$\varsigma_{\mathrm{M}}{}^{n-1/2} = \varsigma_{\mathrm{M}\,\bullet-} \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet-}, \tag{76}$$

$$\boldsymbol{u}^{n-1/2}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet-}, \tag{77}$$

$$u^{n-1/2,\,1}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\bullet}, \tag{78}$$

$$\frac{\partial u^{n-1/2,\,2}_{\mathrm{M}}}{\partial x^1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\bullet}. \tag{79}$$

In what follows, we will split the BVP into three steps [12, 13]:

II.C.4.1 **Approximated velocities.** Proceeding along the lines of the splitting scheme for F of Appendix A 4 b, we introduce the auxiliary velocities $\overline{\boldsymbol{v}}_{\mathrm{M}}$ and $\overline{w}_{\mathrm{M}}$, which approximate, respectively, $\boldsymbol{v}_{\mathrm{M}}{}^n$ and w^n_{M} [13]. We define $\overline{\boldsymbol{v}}_{\mathrm{M}}$ and $\overline{w}_{\mathrm{M}}$ as solutions of the PDEs

$$\begin{aligned}\rho_{\mathrm{M}} \Bigg[\frac{\overline{v}^i_{\mathrm{M}} - v^{n-1,\,i}_{\mathrm{M}}}{\Delta t} + \left(\frac{3}{2} v^{n-1,\,j}_{\mathrm{M}} - \frac{1}{2} v^{n-2,\,j}_{\mathrm{M}} \right) \nabla^{n-1/2}_j V^i_{\mathrm{M}} - 2 V^j_{\mathrm{M}} W_{\mathrm{M}} b^{n-1/2,\,i}{}_j - W_{\mathrm{M}} \nabla^{n-1/2,\,i} W_{\mathrm{M}} \Bigg] =& \\ \nabla^{n-1/2,\,i} \varsigma^*_{\mathrm{M}} + f^i_{\eta_{\mathrm{M}},\mathrm{t}} \left(\boldsymbol{V}_{\mathrm{M}}, W_{\mathrm{M}}, \nu^{n-1/2}, \psi^{n-1/2} \right) +& \\ f^i_{\mathrm{F\,t}}(\varsigma^{\mathrm{r}}_{\mathrm{F}}(\boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1}, \varsigma^{\mathrm{r}}_{\mathrm{F}}{}^{n-3/2}, \boldsymbol{u}^{n-1}_{\mathrm{D}}), \nu^{n-1/2}, \psi^{n-1/2}),&\end{aligned} \tag{80}$$

$$\begin{aligned}\rho_{\mathrm{M}} \left[\frac{\overline{w}_{\mathrm{M}} - w^{n-1}_{\mathrm{M}}}{\Delta t} + V^i_{\mathrm{M}} V^j_{\mathrm{M}} b^{n-1/2}_{ji} + \left(\frac{3}{2} v^{n-1,\,i}_{\mathrm{M}} - \frac{1}{2} v^{n-3/2,\,i}_{\mathrm{M}} \right) \nabla^{n-1/2}_i W_{\mathrm{M}} \right] =& \\ f_{\kappa}(\nu^{n-1/2}, \psi^{n-1/2}) + 2 \varsigma^*_{\mathrm{M}} H^{n-1/2} + f_{\eta_{\mathrm{M}}\,\mathrm{n}}(\boldsymbol{V}_{\mathrm{M}}, W_{\mathrm{M}}, \nu^{n-1/2}, \psi^{n-1/2}) +& \\ f_{\mathrm{F\,n}}(\varsigma^{\mathrm{r}}_{\mathrm{F}}(\boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1}, \varsigma^{\mathrm{r}}_{\mathrm{F}}{}^{n-3/2}, \boldsymbol{u}^{n-1}_{\mathrm{D}}), \nu^{n-1/2}, \psi^{n-1/2}),&\end{aligned} \tag{81}$$

where we have set

$$V^i_{\mathrm{M}} \equiv \frac{\overline{v}^i_{\mathrm{M}} + v^{n-1,\,i}_{\mathrm{M}}}{2}, \tag{82}$$

$$W_{\mathrm{M}} \equiv \frac{\overline{w}_{\mathrm{M}} + w^{n-1}_{\mathrm{M}}}{2}, \tag{83}$$

$$\varsigma^*_{\mathrm{M}} \equiv \varsigma_{\mathrm{M}}{}^{n-3/2}, \tag{84}$$

Equations (80) and (81) are the analog of Eqs. (66) and (67), respectively, with a fixed pressure field ς^*_{M} from the preceding step—in what follows, we will synonymously use the terms ‘pressure’ and ‘tension’ for conciseness [31].

Overall, Eqs. (80) and (81) determine $\overline{v}_{\mathrm{M}}$ and

$\overline{w}_{\rm M}$.

II.C.4.2 **Pressure correction.** Subtracting Eqs. (66) and (80) and neglecting $\mathcal{O}(\Delta t)$ [31], we obtain

$$\rho_{\rm M} \frac{v_{\rm M}^{n,\,i} - \overline{v}_{\rm M}^{i}}{\Delta t} = -\nabla^{n-1/2,\,i} \phi_{\rm M}, \tag{85}$$

where we have defined the pressure increment as

$$\phi_{\rm M} \equiv \varsigma_{\rm M}^{*} - {\varsigma_{\rm M}}^{n-1/2}. \tag{86}$$

Proceeding along the same lines for Eqs. (67) and (81), we obtain

$$w_{\rm M}^{n} = \overline{w}_{\rm M}. \tag{87}$$

By taking the covariant derivative of Eq. (85) and using the mass-conservation equation (65) and Eq. (87), we obtain

$$\nabla_i^{n-1/2} \nabla^{n-1/2,\,i} \phi_{\rm M} = \frac{\rho_{\rm M}}{\Delta t} (\nabla_i^{n-1/2} \overline{v}_{\rm M}^{i} - 2 H^{n-1/2} \overline{w}_{\rm M}). \tag{88}$$

Equation (88) is a second-order PDE which determines $\phi_{\rm M}$, and thus the tension ${\varsigma_{\rm M}}^{n-1/2}$ through Eq. (86).

II.C.4.3 **Velocity correction.** The M velocity ${\boldsymbol{v}_{\rm M}}^{n}$ is determined by means of Eq. (88) in terms of $\overline{\boldsymbol{v}}_{\rm M}$ and $\phi_{\rm M}$.

The BCs for Eqs. (68) to (70), (80), (81), (85), (87) and (88) are

$$\overline{v}_{\rm M}^{i} = 0 \text{ on } \partial\Omega^{\rm r}_{\bullet\!-}, \tag{89}$$

$$\overline{v}_{\rm M}^{i} = 0 \text{ on } \partial\Omega^{\rm r}_{-\!\bullet}, \tag{90}$$

$$\overline{w}_{\rm M} = 0 \text{ on } \partial\Omega^{\rm r}_{\bullet\!-}, \tag{91}$$

$$\nabla_i^{n-1/2} \overline{w}_{\rm M} = 0 \text{ on } \partial\Omega^{\rm r}_{-\!\bullet}, \tag{92}$$

$$\phi_{\rm M} = 0 \text{ on } \partial\Omega^{\rm r}_{\bullet\!-}, \tag{93}$$

$${n_{\rm M}}_i^{n-1/2} \nabla^{n-1/2,\,i} \phi_{\rm M} = 0 \text{ on } \partial\Omega^{\rm r}_{\_}, \tag{94}$$

$$\boldsymbol{u}_{\rm M}^{n-1/2} = 0 \text{ on } \partial\Omega^{\rm r}_{\bullet\!-}, \tag{95}$$

$$u_{\rm M}^{n-1/2,\,1} = 0 \text{ on } \partial\Omega^{\rm r}_{-\!\bullet}, \tag{96}$$

$$\frac{\partial u_{\rm M}^{n-1/2,\,2}}{\partial x^1} = 0 \text{ on } \partial\Omega^{\rm r}_{-\!\bullet}. \tag{97}$$

In Eq. (94), $\boldsymbol{n}_{\rm M}$ is the unit normal to $\partial\Omega_{\_}$: It lies in the tangent bundle of $\Omega_{\_}$, it is directed outside $\Omega_{\_}$, and it is normalized according to ${n_{\rm M}}^i {n_{\rm M}}_i = 1$ [19, 31, 39].

b. Bulk fluid domain The discretization of the equations of motion for D proceeds along the same lines as in Section II A, see Appendix A 4 c. The discrete version of the PDEs (37) and (43) is

$$\frac{\partial S_{{\rm D}\,\alpha\beta}(\boldsymbol{u}_{\rm D}^{n})}{\partial \mathsf{x}_{{\rm r}\,\beta}} = 0 \text{ in } \Omega^{\rm r}, \tag{98}$$

$$\frac{\partial}{\partial \mathsf{x}_{{\rm r}\,\beta}} \frac{{\rm d} S_{{\rm D}\,\alpha\beta}(\boldsymbol{u}_{\rm D}^{n})}{{\rm d}t} = 0 \text{ in } \Omega^{\rm r}, \tag{99}$$

with BCs given by the discrete version of Eqs. (44) to (51):

$$\boldsymbol{u}_{\rm D}^{n} = \boldsymbol{0} \text{ on } \partial\Omega^{\rm r}_{|\square} \cup \partial\Omega^{\rm r}_{\underline{\square}}, \tag{100}$$

$$u_{\rm D}^{n,\,1} = 0 \text{ on } \partial\Omega^{\rm r}_{\square|}, \tag{101}$$

$$\frac{\partial u_{\rm D}^{n,\,2}}{\partial \mathsf{x}_{{\rm r}\,1}} = 0 \text{ on } \partial\Omega^{\rm r}_{\square|}, \tag{102}$$

$$\boldsymbol{u}_{\rm D}^{n} = \boldsymbol{u}_{\rm M}^{n-1/2} \text{ on } \partial\Omega^{\rm r}_{\overline{\square}}, \tag{103}$$

$$\dot{\boldsymbol{u}}_{\rm D}^{n} = \boldsymbol{0} \text{ on } \partial\Omega^{\rm r}_{|\square} \cup \partial\Omega^{\rm r}_{\underline{\square}}, \tag{104}$$

$$\dot{u}_{\rm D}^{n,\,1} = 0 \text{ on } \partial\Omega^{\rm r}_{\square|}, \tag{105}$$

$$\frac{\partial \dot{u}_{\rm D}^{n,\,2}}{\partial \mathsf{x}_{{\rm r}\,1}} = 0 \text{ on } \partial\Omega^{\rm r}_{\square|}, \tag{106}$$

$$\dot{\boldsymbol{u}}_{\rm D}^{n} = \boldsymbol{u}_{\rm M}^{n-1/2} \text{ on } \partial\Omega^{\rm r}_{\overline{\square}}. \tag{107}$$

c. Bulk fluid The discretization of the equations of motion for F proceeds along the same lines as in Section II A, and is split into three steps: Approximated velocity, pressure correction and velocity correction, see Appendix A 4 b.

The resulting PDEs for $\overline{\boldsymbol{v}}_{\rm F}$ and $\varsigma_{\rm F}^{\rm r}$ are

$$\rho_{\rm F} \left\{ \frac{\overline{v}_{\rm F}^{\alpha} - {v_{\rm F}}_{\rm r}^{n-1,\,\alpha}}{\Delta t} + \left[\frac{3}{2} ({v_{\rm F}}_{\rm r}^{n-1,\,\gamma} - \dot{u}_{\rm D}^{n-1,\,\gamma})\, G_{\beta\gamma}(\boldsymbol{u}_{\rm D}^{n-1}) - \frac{1}{2} ({v_{\rm F}}_{\rm r}^{n-2,\,\gamma} - \dot{u}_{\rm D}^{n-2,\,\gamma})\, G_{\beta\gamma}(\boldsymbol{u}_{\rm D}^{n-2}) \right] \frac{\partial V_{\rm F}^{\alpha}}{\partial \mathsf{x}_{{\rm r}\,\beta}} \right\} = G_{\beta\alpha}(\boldsymbol{u}_{\rm D}^{n-1}) \frac{\partial \varsigma_{\rm F}^{{\rm r}\,*}}{\partial \mathsf{x}_{{\rm r}\,\beta}} + \eta_{\rm F}\, G_{\delta\beta}(\boldsymbol{u}_{\rm D}^{n-1}) \frac{\partial}{\partial \mathsf{x}_{{\rm r}\,\delta}} \left(G_{\gamma\beta}(\boldsymbol{u}_{\rm D}^{n-1}) \frac{\partial V_{\rm F}^{\alpha}}{\partial \mathsf{x}_{{\rm r}\,\gamma}} \right), \tag{108}$$

$$\frac{\rho_{\rm F}}{\Delta t} G_{\gamma\alpha}(\boldsymbol{u}_{\rm D}^{n-1}) \frac{\partial \overline{v}_{\rm F}^{\alpha}}{\partial \mathsf{x}_{{\rm r}\,\gamma}} = \tag{109}$$

$$G_{\gamma\alpha}(\boldsymbol{u}_{\rm D}^{n-1}) \frac{\partial}{\partial \mathsf{x}_{{\rm r}\,\gamma}} \left(G_{\beta\alpha}(\boldsymbol{u}_{\rm D}^{n-1}) \frac{\partial \phi_{\rm F}}{\partial \mathsf{x}_{{\rm r}\,\beta}} \right),$$

where we have set

$$\boldsymbol{V}_{\rm F} \equiv \frac{\overline{\boldsymbol{v}}_{\rm F} + {\boldsymbol{v}_{\rm F}}_{\rm r}^{\,n-1}}{2}, \tag{110}$$

$$\varsigma_{\rm F}^{{\rm r}\,*} \equiv \varsigma_{\rm F}^{{\rm r}\,n-3/2}, \tag{111}$$

$$\phi_{\rm F} \equiv \varsigma_{\rm F}^{{\rm r}\,*} - \varsigma_{\rm F}^{{\rm r}\,n-1/2}. \tag{112}$$

The BCs for Eqs. (108) and (109) are [12]

$$\overline{\boldsymbol{v}}_{\rm F} = \boldsymbol{v}_{{\rm F}\,\sqcup} \text{ on } \partial\Omega^{\rm r}_{\sqcup}, \tag{113}$$

$$\overline{\boldsymbol{v}}_{\rm F} = \boldsymbol{0} \text{ on } \partial\Omega^{\rm r}_{\sqsubset}, \tag{114}$$

$$\overline{v}_{\rm F}^{1} = 0 \;\; \text{on } \partial\Omega^{\rm r}_{\sqsupset}, \tag{115}$$

$$G(\boldsymbol{u}_{\rm D}^{n-1})_{\alpha 1} \frac{\partial V_{\rm F}^{2}}{\partial \mathsf{x}_{{\rm r}\,\alpha}} = 0 \;\; \text{on } \partial\Omega^{\rm r}_{\sqsupset}, \tag{116}$$

$$\overline{\boldsymbol{v}}_{\rm F}(x^1, h) = \dot{u}_{\rm M}^{n-1/2}(x^1) \text{ on } \partial\Omega^{\rm r}_{\sqcap}, \tag{117}$$

$$\phi_{\rm F} = 0 \text{ on } \partial\Omega^{\rm r}_{\sqcup}, \tag{118}$$

$$\hat{n}^{{\rm r}\,\gamma} G_{\gamma\alpha}(\boldsymbol{u}_{\rm D}^{n-1}) G_{\beta\alpha}(\boldsymbol{u}_{\rm D}^{n-1}) \frac{\partial \phi_{\rm F}}{\partial \mathsf{x}_{{\rm r}\,\beta}} = \tag{119}$$

$$0 \text{ on } \partial\Omega^{\rm r}_{\sqsubset} \cup \partial\Omega^{\rm r}_{\sqcap},$$

$$G(\boldsymbol{u}_{\rm D}^{n-1})_{\alpha 1} \frac{\partial \phi_{\rm F}}{\partial \mathsf{x}_{{\rm r}\,\alpha}} = 0 \text{ on } \partial\Omega^{\rm r}_{\sqsupset}. \tag{120}$$

Finally, ${\boldsymbol{v}_{\rm F}}_{\rm r}^{\,n}$ is determined from $\overline{\boldsymbol{v}}_{\rm F}$ and $\varsigma_{\rm F}^{{\rm r}\,n-1/2}$ by means of the relation

$$\rho_{\rm F} \frac{\overline{v}_{\rm F}^{\alpha} - {v_{\rm F}}_{\rm r}^{\,n,\,\alpha}}{\Delta t} = G_{\beta\alpha}(\boldsymbol{u}_{\rm D}^{n-1}) \frac{\partial \phi_{\rm F}}{\partial \mathsf{x}_{{\rm r}\,\beta}} \text{ in } \Omega^{\rm r}. \tag{121}$$

Given the M fields $v_{\rm M}^{n-1,i}$, $w_{\rm M}^{n-1,\,i}$, ${\varsigma_{\rm M}}^{n-3/2}$, $\boldsymbol{u}_{\rm M}^{n-3/2}$, $\nu^{n-3/2}$, $\psi^{n-3/2}$, the D fields $\boldsymbol{u}_{\rm D}^{n-1}$, $\dot{\boldsymbol{u}}_{\rm D}^{n-1}$, and the F fields ${\boldsymbol{v}_{\rm F}}_{\rm r}^{\,n-1}$, $\varsigma_{\rm F}^{{\rm r}\,n-3/2}$, we determine the fields at the next time step $n \to n+1$ as follows:

- **Update M.** The M fields $v_{\rm M}^{n,i}$, $w_{\rm M}^{n,\,i}$, ${\varsigma_{\rm M}}^{n-1/2}$, $\boldsymbol{u}_{\rm M}^{n-1/2}$, $\nu^{n-1/2}$, $\psi^{n-1/2}$ are determined by solving the BVP given by Eqs. (68) to (70), (80), (81), (85) and (87) to (97).
- **Update D.** We determine the D displacement field and its time derivative, $\boldsymbol{u}_{\rm D}^{n}$ and $\dot{\boldsymbol{u}}_{\rm D}^{n}$, from the BVPs given by Eqs. (98) to (107).
- **Update F.** We determine the F fields ${\boldsymbol{v}_{\rm M}}^{n}$ and $\varsigma_{\rm F}^{{\rm r}\,n-1/2}$ by solving the BVP given by Eqs. (108), (109) and (113) to (120), and by solving Eq. (121).

The fields ${\boldsymbol{v}_{\rm F}}_{\rm c}^{\,n}$, $\varsigma_{\rm F}^{{\rm c}\,n-1/2}$ in the c configuration are then obtained from the respective fields in the r configuration through Eqs. (5) and (6).

An example of this dynamics is shown in Fig. 6, where we show the interaction between air (F) and an elastomer (M). The channel has dimensions $L = 1\,{\rm m}$, $h = 0.2\,{\rm m}$. Proceeding along the lines of Section II A, M parameters have been estimated by considering the parameters for a three-dimensional material, multiplying them by a unit length corresponding to the out-of-plane depth, and by the M thickness, chosen to be $\Delta = 10^{-3}\,{\rm m}$. The M bending rigidity κ has been estimated from the relation $\kappa = E\Delta^3/[12(1-\nu_{\rm P}^2)]$, which expresses κ as a function of the Young modulus E, the Poisson ratio $\nu_{\rm P}$, and Δ [16]. Typical values for an elastomer are $E \sim 10^7\,{\rm Pa}$ [40] and $\nu_{\rm P} \sim 0.5$ [41], yielding $\kappa = 10^{-3}\,{\rm Kg\,m^3/\,sec^2}$. We chose $\rho_{\rm M} = 1\,{\rm Kg/m}$ and $\eta_{\rm M} = 1\,{\rm Kg\,m/\,sec}$ as the density and viscosity of an elastomer, respectively [42, 43], and we take $\varsigma_{{\rm M}\,\bullet\!-} = 1\,{\rm N}$. The F parameters have been estimated from the air parameters: $\rho_{\rm F} = 1\,{\rm Kg/m^2}$, $\eta_{\rm F} = 10^{-5}\,{\rm Kg/\,sec}$ [23]. Also, we take $\varsigma_{{\rm F}\,\sqcup} = -1\,{\rm N/m^2}$, and $\boldsymbol{v}_{{\rm F}\,\sqcup}^{1} = 0$, $\boldsymbol{v}_{{\rm F}\,\sqcup}^{2}(\mathsf{\mathbf{x}}_{\rm c}) = \frac{3}{2L^2}\mathsf{x}_{{\rm c}\,1}(2L - \mathsf{x}_{{\rm c}\,1})v_0$, where $v_0 = 10^{-2}\,{\rm m/\,sec}$, and $\boldsymbol{v}_{{\rm F}\,\sqcup}^{2}$ corresponds to a Poiseuille flow [18]. Finally, the D exponent is $\zeta = 3$ [15].

III. DISCUSSION

We developed a numerical study on the interaction between a bulk fluid (F) and a Helfrich membrane (M). Our analysis is based on the finite element (FE) method, and it leverages the fi-

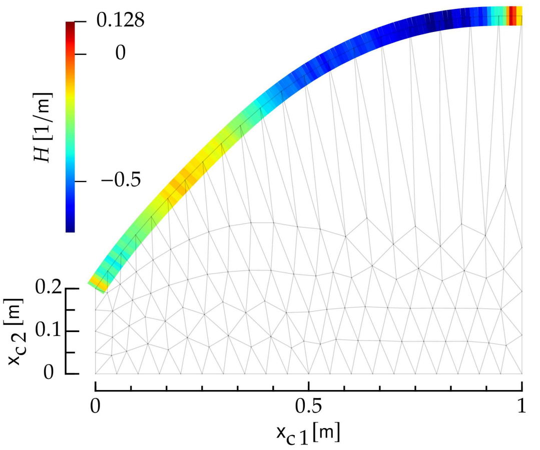

FIG. 7: Curvature of the Helfrich membrane interacting with a bulk fluid shown in Fig. 6. The Figure follows the same notation and it corresponds to the same instant of time as Fig. 6.

nite element computational software (FEniCS) [12], combined with fluId layeR finitE-Element softwarE (IRENE) [31]. In order to guide and introduce the reader to the interaction problem between a bulk fluid and Helfrich membrane, we provided two introductory examples of fluid-structure interaction [15], in which the bulk fluid interacts with a rigid body and with an elastic body, see Section II A and Appendix B. These Sections allow to introduce the necessary physical an mathematical quantities, as well as to prove the stability of our numerical scheme.

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Appendix A: Fluid and rigid body

1. Rigid body motion

In this Section, we will work out the equations of motion for R.

a. *Equation of motion in current-configuration coordinates*

The dynamics of R is given by the equations of motion for a rigid body which, in c coordinates, read [44]

$$I \frac{\mathrm{d}\omega}{\mathrm{d}t} = - \int_{\partial \Omega^{\mathrm{c}}_{\circ}} \mathrm{d}l_{\mathrm{c}}\, \epsilon_{3\alpha\beta}\, \sigma^{\mathrm{c}}_{\mathrm{F}\,\beta\gamma} \hat{n}^{\mathrm{c}\,\gamma}, \tag{A.1}$$

$$\omega \equiv \frac{\mathrm{d}\theta}{\mathrm{d}t}. \tag{A.2}$$

Equation (A.1) relates the angular acceleration of R to the torque of external forces. The moment of inertia of R is denoted by I. The right-hand side (RHS) of Eq. (A.1) contains the line integral along the boundary of R, of the torque of the local force exerted by bulk fluid (F) on R with respect to the left focal point $\boldsymbol{f}$ of R [18], and $\hat{n}^{\mathrm{c}}$ is the normal to $\partial \Omega^{\mathrm{c}}_{\circ}$ pointing towards Ω^{c}.

This force is expressed in terms of the F stress tensor [18] in c coordinates $\sigma^{\mathrm{c}}_{\mathrm{F}}$, given by Eq. (17), where η_{F} and ρ_{F} are the two-dimensional viscosity and density of F, respectively. Also, $\epsilon_{\alpha\beta\gamma}$ is the three-dimensional Levi-Civita symbol. Finally, $\mathrm{d}l_{\mathrm{c}}$ is the line element of $\partial \Omega^{\mathrm{c}}_{\circ}$ in the c configuration, and $\hat{\boldsymbol{n}}^{\mathrm{c}}$ the unit boundary normal to $\partial \Omega^{\mathrm{c}}_{\circ}$ pointing outside Ω^{c}.

b. *Equation of motion in the reference-configuration coordinates*

In r coordinates, Eq. (A.1) reads

$$I \frac{\mathrm{d}\omega}{\mathrm{d}t} = \int_{\partial \Omega^{\mathrm{r}}_{\circ}} \mathrm{d}s\, \epsilon_{\alpha\beta} \left[\varphi^{\alpha,\,t}(\boldsymbol{\xi}(s)) - f^{\alpha} \right] \sigma^{\mathrm{r}}_{\mathrm{F}\,\beta\gamma}(\boldsymbol{\xi}(s)) \epsilon_{\gamma\lambda} F_{\lambda\mu}(\boldsymbol{u}^{t}_{\mathrm{D}}(\boldsymbol{\xi}(s))) \frac{\mathrm{d}\xi^{\mu}(s)}{\mathrm{d}s}. \tag{A.3}$$

In Eq. (A.3), the integral in the RHS is over the curvilinear boundary $\partial\Omega^{\mathrm{r}}_{\bigcirc}$, which is parametrized as $\mathsf{x}_{\mathrm{r}} = \boldsymbol{\xi}(s)$, where s is the curvilinear coordinate. Also, $\epsilon_{\alpha\beta}$ is the two-dimensional Levi-Civita symbol [19], and $\sigma^{\mathrm{r}}_{\mathrm{F}}$ denotes the stress tensor in the r configuration, which is given by Eq. (71), where G is given by Eq. (62).

2. Fluid motion

In this Section, we will work out the equations of motion for F.

a. *Equations of motion in the current-configuration coordinates*

The equations of motion, expressed in terms of the c fields $v_{\mathrm{F}\,\mathrm{c}}$ and $\varsigma^{\mathrm{c}}_{\mathrm{F}}$, are Eqs. (52) and (53), see Section II C 3 for details.

The boundary conditions (BCs) for Eqs. (52), (53), (A.1) and (A.2) are

$$v_{\mathrm{F}\,\mathrm{c}}^{\,1}(\mathsf{x}_{\mathrm{c}}) = \boldsymbol{v}_{\mathrm{F}\,\sqsubset} \quad \text{on } \partial\Omega^{\mathrm{c}}_{\sqsubset}, \tag{A.4}$$
$$v_{\mathrm{F}\,\mathrm{c}}^{\,2}(\mathsf{x}_{\mathrm{c}}) = 0 \quad \text{on } \partial\Omega^{\mathrm{c}}_{\sqsubset}, \tag{A.5}$$
$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}(\mathsf{x}_{\mathrm{c}}) = \mathbf{0} \quad \text{on } \partial\Omega^{\mathrm{c}}_{\underline{\square}}, \tag{A.6}$$
$$v_{\mathrm{F}\,\mathrm{c}}^{\,\alpha}(\mathsf{x}_{\mathrm{c}}) = \frac{\mathrm{d}[R(\theta)]_{\alpha\beta}}{\mathrm{d}t}[\psi^{t\,\beta}(\mathsf{x}_{\mathrm{c}}) - f^{\beta}] \quad \text{on } \partial\Omega^{\mathrm{c}}_{\bigcirc}, \tag{A.7}$$
$$\eta_{\mathrm{F}}\frac{\partial v_{\mathrm{F}\,\mathrm{c}}^{\,\alpha}}{\partial \mathsf{x}_{\mathrm{c}\,1}} = 0 \quad \text{on } \partial\Omega^{\mathrm{c}}_{\sqsupset}, \tag{A.8}$$
$$\varsigma^{\mathrm{c}}_{\mathrm{F}}(\mathsf{x}_{\mathrm{c}}) = 0 \quad \text{on } \partial\Omega^{\mathrm{c}}_{\sqsupset}. \tag{A.9}$$

Equations (A.4) and (A.5) ensure that F is injected on the boundary $\partial\Omega^{\mathrm{c}}_{\sqsubset}$ in a direction parallel to the $\mathsf{x}_{\mathrm{c}\,1}$ axis, and Eq. (A.6) is a no-slip BC [18] along the channel walls $\partial\Omega^{\mathrm{c}}_{\underline{\square}}$, see Fig. 2. Equation (A.7) ensures that, at the boundary $\partial\Omega^{\mathrm{c}}_{\bigcirc}$, the F velocity matches the rotational velocity of R. Given that R can only rotate about its left focal point $\boldsymbol{f}$, the R rotational velocity of a point $\mathsf{x}_{\mathrm{c}} \in \partial\Omega^{\mathrm{c}}_{\bigcirc}$ is given by

$$\frac{\mathrm{d}[R(\theta)]_{\alpha\beta}}{\mathrm{d}t}[\psi^{t\,\beta}(\mathsf{x}_{\mathrm{c}}) - c_{\beta}], \tag{A.10}$$

where $\boldsymbol{R}$ is the rotation matrix corresponding to a counterclockwise rotation:

$$\boldsymbol{R}(\theta) \equiv \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{A.11}$$

Finally, Eqs. (A.8) and (A.9) ensure, respectively, that on $\partial\Omega^{\mathrm{c}}_{\sqsupset}$ there is zero traction and tension; this corresponds to the hypothesis that there is free flow at $\partial\Omega^{\mathrm{c}}_{\sqsupset}$ [12, 18].

b. *Equations of motion in the reference-configuration coordinates*

We will now rewrite the equations of motion and the BCs in terms of $\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}$, $\varsigma^{\mathrm{r}}_{\mathrm{F}}$, and the r coordinate x_{r}. The equations of motion (52) and (53) yield, respectively, Eqs. (60) and (61), see Section II C 3 b. The BCs (A.4) to (A.9) yield as

$$v_{\mathrm{F}\,\mathrm{r}}^{\,1}(\mathsf{x}_{\mathrm{r}}) = \boldsymbol{v}_{\mathrm{F}\,\sqsubset} \quad \text{on } \partial\Omega^{\mathrm{r}}_{\sqsubset}, \tag{A.12}$$
$$v_{\mathrm{F}\,\mathrm{r}}^{\,2}(\mathsf{x}_{\mathrm{r}}) = 0 \quad \text{on } \partial\Omega^{\mathrm{r}}_{\sqsubset}, \tag{A.13}$$
$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}(\mathsf{x}_{\mathrm{r}}) = \mathbf{0} \quad \text{on } \partial\Omega^{\mathrm{r}}_{\underline{\square}}, \tag{A.14}$$
$$v_{\mathrm{F}\,\mathrm{r}}^{\,\alpha}(\mathsf{x}_{\mathrm{r}}) = \frac{\mathrm{d}[R(\theta)]_{\alpha\beta}}{\mathrm{d}t}(\mathsf{x}_{\mathrm{r}\,\beta} - f^{\beta}) \quad \text{on } \partial\Omega^{\mathrm{r}}_{\bigcirc}, \tag{A.15}$$
$$\eta_{\mathrm{F}} G_{\beta 1}(\boldsymbol{u}^{t}_{\mathrm{D}}(\mathsf{x}_{\mathrm{r}}))\frac{\partial v_{\mathrm{F}\,\mathrm{r}}^{\,\alpha}}{\partial \mathsf{x}_{\mathrm{r}\,\beta}} = 0 \quad \text{on } \partial\Omega^{\mathrm{r}}_{\sqsupset}, \tag{A.16}$$
$$\varsigma^{\mathrm{r}}_{\mathrm{F}}(\mathsf{x}_{\mathrm{r}}) = 0 \quad \text{on } \partial\Omega^{\mathrm{r}}_{\sqsupset}. \tag{A.17}$$

3. Motion of bulk fluid domain

The equations of motion (60), (61), (A.3) and (A.12) to (A.17) involve the displacement field $\boldsymbol{u}_{\mathrm{D}}$, which describes how the region—or domain—which contains F, is deformed in time, cf. Fig. 2. We will denote this region as the D. We observe that this field has, so far, been left arbitrary, and we have the freedom to choose it.

Here, $\boldsymbol{u}_{\mathrm{D}}$ will be determined with an analogy with the theory of elasticity [15]. We imagine that, as R rotates about its focal point, each point $\mathsf{x}_{\mathrm{r}} \in \Omega^{\mathrm{r}}$ moves into a point $\mathsf{x}_{\mathrm{c}} \in \Omega^{\mathrm{c}}$, and that the domain Ω^{r} is deformed, e.g., locally stretched and compressed, into Ω^{c}, like an elastic medium. We will determine $\boldsymbol{u}_{\mathrm{D}}$ as the solution of a boundary-value problem (BVP) given by the elastic model (37), see Section II C 2 for details.

The BCs for the partial differential equation (PDE) (37) are

$$\boldsymbol{u}_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\blacksquare}, \tag{A.18}$$
$$\boldsymbol{u}_{\mathrm{D}} = \boldsymbol{f} - \mathsf{x}_{\mathrm{r}} + \boldsymbol{R}(\theta)\cdot(\mathsf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\bigcirc}. \tag{A.19}$$

Equation (A.18) enforces the fact that the mesh is not deformed at the boundary $\partial\Omega^{\mathrm{r}}_{\blacksquare}$. Finally, Eq. (A.19) ensures that the mesh deformation at $\partial\Omega^{\mathrm{r}}_{\mathbf{O}}$ matches the deformation induced by the rotation of R.

Equations (37), (A.18) and (A.19) constitute the BVP which determines $\boldsymbol{u}_{\mathrm{D}}$. Along the lines of BVPs in the theory of elasticity, such BVP is already, naturally formulated in the r configuration [16].

From the BVP above, we obtain a BVP for $\dot{\boldsymbol{u}}_{\mathrm{D}}$, which will be needed in the following to describe the D motion. By deriving both sides of Eqs. (37), (A.18) and (A.19), we obtain, respectively, Eq. (43) and

$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{A.20}$$

$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \omega \frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta} \cdot (\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\mathbf{O}}, \tag{A.21}$$

where in Eqs. (A.20) and (A.21) we used the fact that $\mathbf{x}_{\mathrm{r}}$ is independent of time, and in Eq. (A.21) we substituted Eq. (A.2). The left-hand side (LHS) of Eq. (43) depends on the second partial derivatives of $\boldsymbol{u}_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}$—see Eqs. (38) to (42). As a result, given θ, ω, and $\boldsymbol{u}_{\mathrm{D}}$ from Eqs. (37), (A.18) and (A.19), the BVP (43), (A.20) and (A.21) determines $\dot{\boldsymbol{u}}_{\mathrm{D}}$.

Overall Eqs. (37), (43), (60), (61), (A.2), (A.3) and (A.12) to (A.21) constitute a BVP for $\theta, \omega, \boldsymbol{v}_{\mathrm{F\,r}}, \varsigma^{\mathrm{r}}_{\mathrm{F}}$, $\boldsymbol{u}_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}$ which, combined with the temporal BCs

$$\theta(t=0) = \theta_0 \tag{A.22}$$

$$\omega(t=0) = \omega_0 \tag{A.23}$$

$$\boldsymbol{v}_{\mathrm{F\,r}}(\mathbf{x}_{\mathrm{r}}, t=0) = \boldsymbol{v}_{\mathrm{F\,r}\,0}(\mathbf{x}_{\mathrm{r}}), \tag{A.24}$$

$$\varsigma^{\mathrm{r}}_{\mathrm{F}}(\mathbf{x}_{\mathrm{r}}, t=0) = \varsigma^{\mathrm{r}}_{\mathrm{F}\,0}(\mathbf{x}_{\mathrm{r}}), \tag{A.25}$$

determine the dynamics of the system.

4. Temporal discretization

To numerically solve for the dynamics in a time span $0 \le t \le T$, we discretize time by setting $\Delta t \equiv T/N$, $t_n \equiv n\,\Delta t$, with $n = 0, 1, \cdots, N$.

We combine the Crank Nicolson (CN) discretization method for Navier-Stokes (NS) equations [12, 38] with the arbitrary Lagrangian-Eulerian (ALE) formulation, by discretizing the dynamical equations for R, F and D—see below.

a. *Rigid body*

Proceeding along the lines of the CN scheme [38], we define fields at staggered, integer and semi-integer time steps [12]—see for example Eqs. (63) and (64) for $\boldsymbol{v}_{\mathrm{F\,r}}$ and $\varsigma^{\mathrm{r}}_{\mathrm{F}}$—and similarly for other quantities. We discretize Eqs. (A.2) and (A.3) and, neglecting $\mathcal{O}(\Delta t)$, we obtain

$$\frac{\theta^n - \theta^{n-1}}{\Delta t} = \omega^n, \tag{A.26}$$

$$\begin{aligned}\frac{\omega^n - \omega^{n-1}}{\Delta t} =& I \int_{\partial\Omega^{\mathrm{r}}_{\mathbf{O}}} \mathrm{d}s\, \epsilon_{\alpha\beta} \left[\xi^\alpha(s) + u^{n-1,\,\alpha}(\boldsymbol{\xi}(s)) - f^\alpha\right] \sigma^{\mathrm{r}\;n-1}_{\mathrm{F}\,\beta\gamma}(\boldsymbol{\xi}(s); \boldsymbol{v}^{n-1}_{\mathrm{F\,r}}, \varsigma^{\mathrm{r}\;n-3/2}_{\mathrm{F}}) \times \\ & \epsilon_{\gamma\lambda} F_{\lambda\mu}(\boldsymbol{u}^{n-1}_{\mathrm{D}}(\boldsymbol{\xi}(s))) \frac{\mathrm{d}\xi^\mu(s)}{\mathrm{d}s},\end{aligned} \tag{A.27}$$

where in Eq. (A.27) we used Eqs. (1) and (3) and we explicitly indicated the dependence of the stress tensor (71) on $\boldsymbol{v}_{\mathrm{F\,c}}$ and $\varsigma^{\mathrm{c}}_{\mathrm{F}}$.

b. *Bulk fluid*

To discretize in time the F problem, we will apply the CN splitting scheme [12, 38] to the NS equations (60) and (61) in the r configurations. This splitting scheme is derived form the incremental pressure correction scheme [45], in which the solution of the NS equations is split into three intermediate steps. In what follows, we will synonymously use the terms 'pressure' and 'surface tension' for conciseness [31].

The discrete form of the PDEs Eqs. (60) and (61) is

$$\rho_{\mathrm{F}}\left[\frac{{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha}-{v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{\Delta t}+\frac{3}{2}\left({v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\gamma}-\dot{u}^{n-1,\,\gamma}\right)G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{\partial \mathsf{x}_{\mathrm{r}\,\beta}}-\frac{1}{2}\left({v_{\mathrm{F}}}_{\mathrm{r}}^{n-2,\,\gamma}-\dot{u}^{n-2,\,\gamma}\right)\times \right. \tag{A.28}$$

$$\left. G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{n-2})\frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{n-2,\,\alpha}}{\partial \mathsf{x}_{\mathrm{r}\,\beta}}\right]=$$

$$G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial {\varsigma_{\mathrm{F}}^{\mathrm{r}}}^{\,n-1/2}}{\partial \mathsf{x}_{\mathrm{r}\,\beta}}+\eta_{\mathrm{F}}\,G_{\delta\beta}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}\,\delta}}\left[G_{\gamma\beta}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}\,\gamma}}\left(\frac{{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha}+{v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{2}\right)\right],$$

$$G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha}}{\partial \mathsf{x}_{\mathrm{r}\,\beta}}=0, \tag{A.29}$$

where we omitted the dependence on x_{r} for clarity. Also, in the LHS of Eq. (A.28) we rewrote the convective term according to the Adams-Bashforth discretization scheme [12, 46], and we used Eqs. (1) and (3) to rewrite the time derivative of $\boldsymbol{\varphi}$ in terms of $\dot{\boldsymbol{u}}_{\mathrm{D}}$.

The discrete form of the BCs (A.12) to (A.17) reads

$$\begin{aligned}
{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,1} &= \boldsymbol{v}_{\mathrm{F}\,\square} \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A.30)}\\
{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,2} &= 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A.31)}\\
{\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{\,n} &= \mathbf{0} \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A.32)}\\
{\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}} &= \omega^{n}\,\left.\frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta}\right|_{\theta=\theta_n}\cdot(\mathsf{x}_{\mathrm{r}}-\boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, && \text{(A.33)}\\
\eta_{\mathrm{F}}\,G_{\beta 1}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}\,\beta}}\left(\frac{{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha}+{v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{2}\right) &= 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A.34)}\\
{\varsigma_{\mathrm{F}}^{\mathrm{r}}}^{\,n-1/2} &= 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}. && \text{(A.35)}
\end{aligned}$$

We will now split the BVP into three steps [12, 13]:

1. **Approximated velocity.** We introduce the velocity $\overline{\boldsymbol{v}}_{\mathrm{F}}$, which approximates the true velocity ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}$, and which satisfies the BVP given by Eq. (108) and BCs

$$\begin{aligned}
\overline{v}_{\mathrm{F}}^{1} &= \boldsymbol{v}_{\mathrm{F}\,\square} \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A.36)}\\
\overline{v}_{\mathrm{F}}^{2} &= 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A.37)}\\
\overline{\boldsymbol{v}}_{\mathrm{F}} &= \mathbf{0} \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A.38)}\\
\overline{\boldsymbol{v}}_{\mathrm{F}} &= \omega^{n}\,\left.\frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta}\right|_{\theta=\theta_n}\cdot(\mathsf{x}_{\mathrm{r}}-\boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, && \text{(A.39)}\\
\eta_{\mathrm{F}}\,G_{\beta 1}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial V_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}\,\beta}} &= 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}. && \text{(A.40)}
\end{aligned}$$

where $\boldsymbol{V}_\mathrm{F}$ and $\varsigma_\mathrm{F}^{\mathrm{r}\,*}$ are given by Eqs. (110) and (111).

2. **Pressure correction.**

   Subtracting Eqs. (108) and (A.28) and neglecting $\mathcal{O}(\Delta t)$, we obtain Eq. (121), where the surface-tension increment ϕ_F is defined by Eq. (112). By applying $G_{\gamma\alpha}(\boldsymbol{u}_\mathrm{D}^{n-1})\partial/\partial \mathsf{x}_{\mathrm{r}\,\gamma}$ to both sides of Eq. (121) and using Eq. (A.29), we obtain the second-order PDE (109) for ϕ_F.

   We will now work out the BCs for Eq. (109). Equations (111), (112) and (A.35) imply the BC

$$\phi_\mathrm{F} = 0 \text{ on } \partial\Omega^\mathrm{r}_{\shortmid}. \tag{A.41}$$

   Multiplying Eq. (121) by $G_{\gamma\alpha}(\boldsymbol{u}_\mathrm{D}^{n-1})\hat{n}^{\mathrm{r}\,\gamma}$, where $\hat{\boldsymbol{n}}^\mathrm{r}$ is the unit boundary normal in the r configuration, we obtain

$$\frac{\rho_\mathrm{F}}{\Delta t} G_{\gamma\alpha}(\boldsymbol{u}_\mathrm{D}^{n-1})\hat{n}^{\mathrm{r}\,\gamma}(\overline{v}_\mathrm{F}^{\alpha} - v_{\mathrm{F}_\mathrm{r}}^{\;n,\,\alpha}) = \hat{n}^{\mathrm{r}\,\gamma} G_{\gamma\alpha}(\boldsymbol{u}_\mathrm{D}^{n-1}) G_{\beta\alpha}(\boldsymbol{u}_\mathrm{D}^{n-1}) \frac{\partial \phi_\mathrm{F}}{\partial \mathsf{x}_{\mathrm{r}\,\beta}}. \tag{A.42}$$

   Combining Eq. (A.42) with Eqs. (A.30) to (A.33) and (A.36) to (A.39), we obtain the second BC

$$\hat{n}^{\mathrm{r}\,\gamma} G_{\gamma\alpha}(\boldsymbol{u}_\mathrm{D}^{n-1}) G_{\beta\alpha}(\boldsymbol{u}_\mathrm{D}^{n-1}) \frac{\partial \phi_\mathrm{F}}{\partial \mathsf{x}_{\mathrm{r}\,\beta}} = 0 \text{ on } \partial\Omega^\mathrm{r}_{\square} \cup \partial\Omega^\mathrm{r}_{-} \cup \partial\Omega^\mathrm{r}_{\circ}. \tag{A.43}$$

   The BVP (109), (A.41) and (A.43) determines ϕ_F and, through the definition (112), the surface tension $\varsigma_\mathrm{c}^{\;n-1/2}$.

3. **Velocity correction.** The velocity $\boldsymbol{v}_{\mathrm{F}_\mathrm{r}}^{\;n}$ is determined from the approximate velocity $\overline{\boldsymbol{v}}_\mathrm{F}$ from Eq. (121).

c. *Bulk fluid domain*

The discrete version of the BVP for $\boldsymbol{u}_\mathrm{D}$, Eqs. (37), (A.18) and (A.19), is given by Eq. (98) and

$$\boldsymbol{u}_\mathrm{D}^n = \mathbf{0} \text{ on } \partial\Omega^\mathrm{r}_{\blacksquare}, \tag{A.44}$$
$$\boldsymbol{u}_\mathrm{D}^n = \boldsymbol{f} - \mathsf{x}_\mathrm{r} + \boldsymbol{R}(\theta^n)\cdot(\mathsf{x}_\mathrm{r} - \boldsymbol{f}) \text{ on } \partial\Omega^\mathrm{r}_{\circ}. \tag{A.45}$$

The discrete version of the BVP for $\dot{\boldsymbol{u}}_\mathrm{D}$, Eqs. (43), (A.20) and (A.21), is given by Eq. (99) and

$$\dot{\boldsymbol{u}}_\mathrm{D}^n = \mathbf{0} \text{ on } \partial\Omega^\mathrm{r}_{\blacksquare}, \tag{A.46}$$

$$\dot{\boldsymbol{u}}_\mathrm{D}^n = \omega \frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta}\cdot(\mathsf{x}_\mathrm{r} - \boldsymbol{f}) \text{ on } \partial\Omega^\mathrm{r}_{\circ}. \tag{A.47}$$

We can now iterate in time for the ensemble of fields of the F-R system as follows. At each step n, given θ^{n-1}, ω^{n-1}, $\boldsymbol{v}_{\mathrm{F}_\mathrm{r}}^{\;n-1}$, $\varsigma_\mathrm{F}^{\mathrm{r}\,n-3/2}$, $\boldsymbol{u}_\mathrm{D}^{n-1}$ and $\dot{\boldsymbol{u}}_\mathrm{D}^{n-1}$ from the preceding step, we

1. **Update R.** Solve the algebraic equations Eqs. (A.26) and (A.27) for θ^n and ω^n.
2. **Update D.** Solve the BVPs (98), (99) and (A.44) to (A.47) for $\boldsymbol{u}_\mathrm{D}^n$ and $\dot{\boldsymbol{u}}_\mathrm{D}^n$.
3. **Update F.** Solve the BVPs (108) and (A.36) to (A.40) and (A.41) to (A.43) for $\overline{\boldsymbol{v}}_\mathrm{F}$ and $\varsigma_\mathrm{F}^{\mathrm{r}\,n-1/2}$, and obtain $\boldsymbol{v}_{\mathrm{F}_\mathrm{r}}^{\;n}$ from Eq. (121).

The fields $\boldsymbol{v}_{\mathrm{F}_\mathrm{c}}^{\;n-1}$, $\varsigma_\mathrm{F}^{\mathrm{c}\,n-3/2}$ in the c configuration are then obtained from the respective fields in the r configuration through Eqs. (5) and (6).

5. Characteristic time scale

As shown in Fig. A.1, the dynamics of Fig. 1 involves a characteristic time scales τ_B, i.e., the period of the angular oscillations of B. This time scale can be estimated by considering the dynamical equation (A.1) for B, and approximating its LHS as I/τ_B^2 and its RHS as $(\eta_\mathrm{F} v_0/a\, 2\pi a)a$. Here, $\eta_\mathrm{F} v_0/a$ is the viscous component of the force per unit length exerted by F on B, cf. Eq. (17), $2\pi a$ estimates the total length of the boundary of B, and a approximates the lever arm of the force above. Putting everything together, we obtain

$$\tau_\mathrm{B} \sim \sqrt{\frac{I}{2\pi a \eta_\mathrm{F} v_0}}. \tag{A.48}$$

For the parameters of Fig. 1, Eq. (A.48) yields $\tau_\mathrm{B} \sim 130$ sec, which is in agreement with the period of the oscillations in Fig. A.1.

Appendix B: Fluid and elastic body

1. Equations of motion

The equations of motion for the dynamics of F and elastic body (E) are the following.

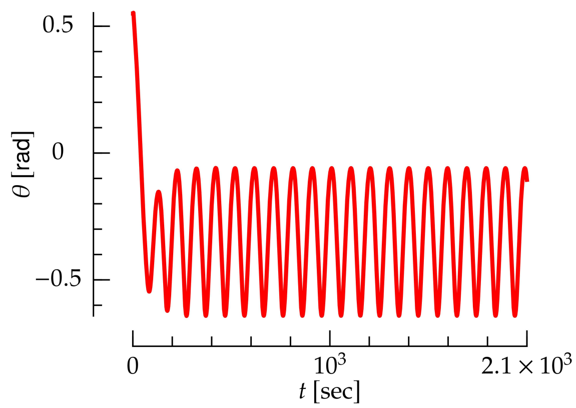

FIG. A.1: Angle of the rigid body as a function of time for the interaction between a bulk fluid and a rigid body shown in Fig. 1.

First, the dynamics of F is governed by the NS and mass-conservation equations (52) and (53), respectively. Their BCs are Eqs. (A.4) to (A.6), (A.8) and (A.9), which stay unchanged with respect to Section II A. On the other hand, BC (A.7), which ensures that the F velocity matches the B velocity at the interface between F and B, now reads

$$\boldsymbol{v}_{\mathrm{F\,c}}(\mathsf{x}_{\mathrm{c}}) = \left.\frac{\mathrm{d}\boldsymbol{u}_{\mathrm{E}}(\mathsf{x}_{\mathrm{r}})}{\mathrm{d}t}\right|_{\mathsf{x}_{\mathrm{r}}=\psi^t(\mathsf{x}_{\mathrm{c}})} \text{ on } \partial\Omega^{\mathrm{c}}_{\bigcirc} \tag{B.1}$$

Second, the motion of E is governed by Newton's equations of motion for an elastic body [16]

$$\rho_{\mathrm{E}}\frac{\mathrm{d}^2 u^{\alpha}_{\mathrm{E}}(\mathsf{x}_{\mathrm{r}})}{\mathrm{d}t^2} = \frac{\partial S_{\mathrm{E}\,\alpha\beta}(\boldsymbol{u}_{\mathrm{E}})}{\partial \mathsf{x}_{\mathrm{r}\,\beta}}, \tag{B.2}$$

where ρ_{E} is the density of E, which is assumed to be independent of space. Also, $\boldsymbol{u}_{\mathrm{E}}$ is the deformation field of E, which is defined along the lines of Eq. (1). The E stress tensor is derived from a compressible neo-Hookean model [47], and it reads

$$S_{\mathrm{E}\alpha\beta} \equiv \frac{\mu}{\det F}\left(-\frac{1}{2}C_{\gamma\gamma}G_{\alpha\beta} + F_{\beta\alpha}\right) + K[(\det(F))^2 - 1]G_{\alpha\beta}, \tag{B.3}$$

where

$$C_{\alpha\beta} \equiv \delta_{\alpha\beta} + 2E_{\alpha\beta}. \tag{B.4}$$

We have chosen this model because it is stable under compression, i.e., its potential energy diverges as $\det F \to 0$ [47]. As opposed to linear models [16] and to the elastic model of Appendix A 3, this model proves to yield a stable dynamics for E as it is compressed by F, see below and Fig. 4. The BCs for Eq. (B.2) are

$$\boldsymbol{u}_{\mathrm{E}}(\mathsf{x}_{\mathrm{r}}) = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\odot}, \tag{B.5}$$

$$\begin{aligned} S_{\mathrm{E}\alpha\beta}(\boldsymbol{u}^t_{\mathrm{E}})\big|_{\mathsf{x}_{\mathrm{r}}=\boldsymbol{\xi}(s)}\,\epsilon_{\beta\gamma}\frac{\mathrm{d}\xi^{\gamma}}{\mathrm{d}s} &= \\ \sigma^{\mathrm{c}}_{\mathrm{F}\,\alpha\beta}(\boldsymbol{\varphi}^t(\boldsymbol{\xi}(s)))\epsilon_{\beta\gamma}\frac{\mathrm{d}\varphi^{t,\,\gamma}(\boldsymbol{\xi}(s))}{\mathrm{d}s} &\text{ on } \partial\Omega^{\mathrm{r}}_{\bigcirc}. \end{aligned} \tag{B.6}$$

In the LHS of Eq. (B.6), the quantity $\epsilon_{\beta\gamma}\frac{\mathrm{d}\xi^{\gamma}}{\mathrm{d}s}$ represents the unit normal to $\partial\Omega^{\mathrm{r}}_{\bigcirc}$, and similarly for the RHS.

Finally, the equations of motion for D are (37) and (43), with BCs (A.18) and (A.20) and

$$\boldsymbol{u}^t_{\mathrm{D}}(\mathsf{x}_{\mathrm{r}}) = \boldsymbol{u}^t_{\mathrm{E}}(\mathsf{x}_{\mathrm{r}}) \text{ on } \partial\Omega^{\mathrm{r}}_{\bigcirc}, \tag{B.7}$$

$$\dot{\boldsymbol{u}}^t_{\mathrm{D}}(\mathsf{x}_{\mathrm{r}}) = \dot{\boldsymbol{u}}^t_{\mathrm{E}}(\mathsf{x}_{\mathrm{r}}) \text{ on } \partial\Omega^{\mathrm{r}}_{\bigcirc}. \tag{B.8}$$

Equation (B.7) and Eq. (B.8) ensure that the displacement and velocity of E and D are conforming at the E-D interface, and they replace Eqs. (A.19) and (A.21), respectively.

2. Dynamics

The system dynamics is obtained as follows: given $\boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1}$, $\varsigma^{\mathrm{r}\,n-3/2}_{\mathrm{F}}$, $\boldsymbol{u}^{n-1}_{\mathrm{E}}$, $\dot{\boldsymbol{u}}^{n-1}_{\mathrm{E}}$, $\dot{\boldsymbol{u}}^{n-1}_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}^{n-1}_{\mathrm{D}}$ from the preceding step, we

- **Update E.** Solve for $\boldsymbol{u}^n_{\mathrm{E}}$ and $\dot{\boldsymbol{u}}^n_{\mathrm{E}}$ the discrete version of Eq. (B.2)

$$\rho_{\mathrm{E}}\frac{\dot{u}^{n,\,\alpha}_{\mathrm{E}} - \dot{u}^{n-1,\,\alpha}_{\mathrm{E}}}{\Delta t} = \frac{\partial S_{\mathrm{E}\alpha\beta}(\boldsymbol{u}^n_{\mathrm{E}})}{\partial \mathsf{x}_{\mathrm{r}\,\beta}}, \tag{B.9}$$

$$\frac{\boldsymbol{u}^n_{\mathrm{E}} - \boldsymbol{u}^{n-1}_{\mathrm{E}}}{\Delta t} = \dot{\boldsymbol{u}}_{\mathrm{E}}, \tag{B.10}$$

with BCs obtained from Eqs. (B.5) and (B.6):

$$\boldsymbol{u}^n_{\mathrm{E}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\odot}, \tag{B.11}$$

$$\epsilon_{\beta\gamma} \; S_{\mathrm{E}\alpha\beta}(\boldsymbol{u}^n_{\mathrm{E}})\big|_{\mathbf{x}_{\mathrm{r}}=\boldsymbol{\xi}(s)} \frac{\mathrm{d}\xi^\gamma}{\mathrm{d}s} = \tag{B.12}$$

$$\sigma^{\mathrm{r}}_{\mathrm{F}\alpha\beta}(\boldsymbol{\xi}(s); \boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}{}^{n-1}, \varsigma^{\mathrm{r}\,n-3/2}_{\mathrm{F}}, \boldsymbol{u}^n_{\mathrm{E}})\epsilon_{\beta\gamma} \; F_{\gamma\delta}(\boldsymbol{u}^{n-1}_{\mathrm{E}})\big|_{\mathbf{x}_{\mathrm{r}}=\boldsymbol{\xi}(s)} \frac{\mathrm{d}\xi^\delta}{\mathrm{d}s} \text{ on } \partial\Omega^{\mathrm{r}}_{\mathbf{O}}.$$

In Eq. (B.12), we have used Eqs. (2) and (3) to rewrite the derivative of $\boldsymbol{\phi}(\boldsymbol{\xi}(s))$.

- **Update D.** Solve for $\boldsymbol{u}^n_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}^n_{\mathrm{D}}$ the BVPs given by Eqs. (98) and (99) with BCs (A.44), (A.46) and

$$\boldsymbol{u}^n_{\mathrm{D}} = \boldsymbol{u}^n_{\mathrm{E}} \text{ on } \partial\Omega^{\mathrm{r}}_{\mathbf{O}}, \tag{B.13}$$

$$\dot{\boldsymbol{u}}^n_{\mathrm{D}} = \dot{\boldsymbol{u}}^n_{\mathrm{E}} \text{ on } \partial\Omega^{\mathrm{r}}_{\mathbf{O}}. \tag{B.14}$$

- **Update F.** Solve the BVPs given by (108) and (109) and BCs (A.36) to (A.38), (A.40), (A.41) and (A.43) and

$$\overline{\boldsymbol{v}}_{\mathrm{F}} = \dot{\boldsymbol{u}}^n_{\mathrm{E}} \text{ on } \partial\Omega^{\mathrm{r}}_{\mathbf{O}}, \tag{B.15}$$

which replaces Eq. (A.40) for a rigid body, and solve Eq. (121).

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